



Scuola Superiore
Sant'Anna

Master of Science in Bionics Engineering

Academic Year 2025-2026

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Bioinspired and Soft Robotics

Soft Robotics Technologies

Jamming Transition

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Definition

JSEL The First Steps of a Robot Based on Jamming Skin Enabled Locomotion

Annan Mozeika
Erik Steltz
Heinrich Jaeger

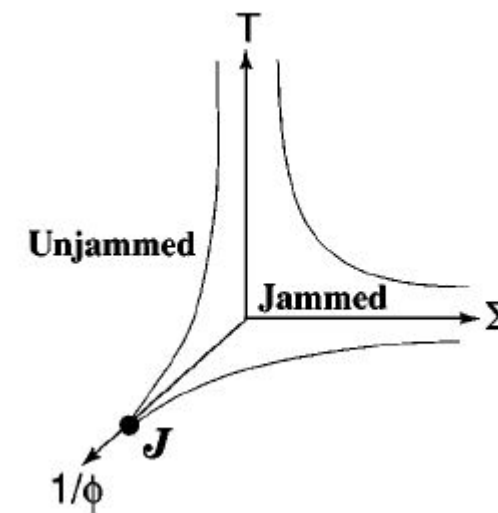
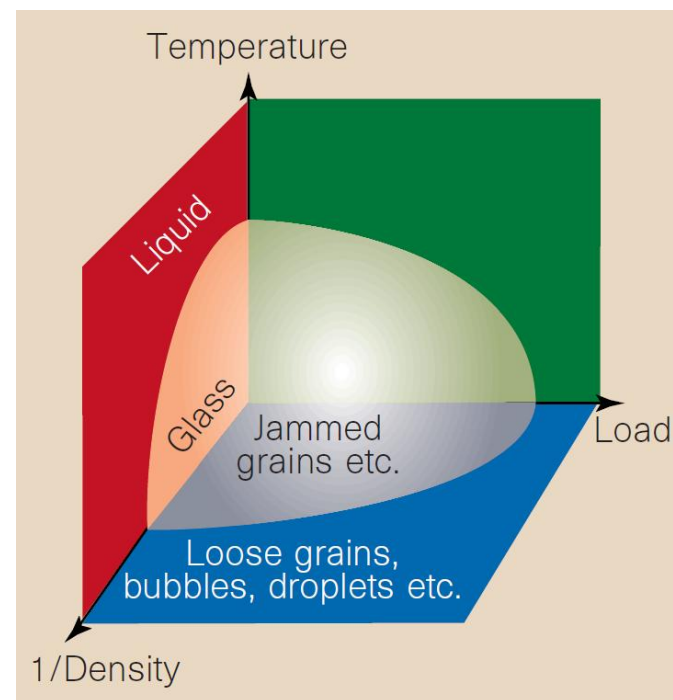


Funded under the DARPA Chemical Robots program, contract W911NF-08-1-0209



Definition

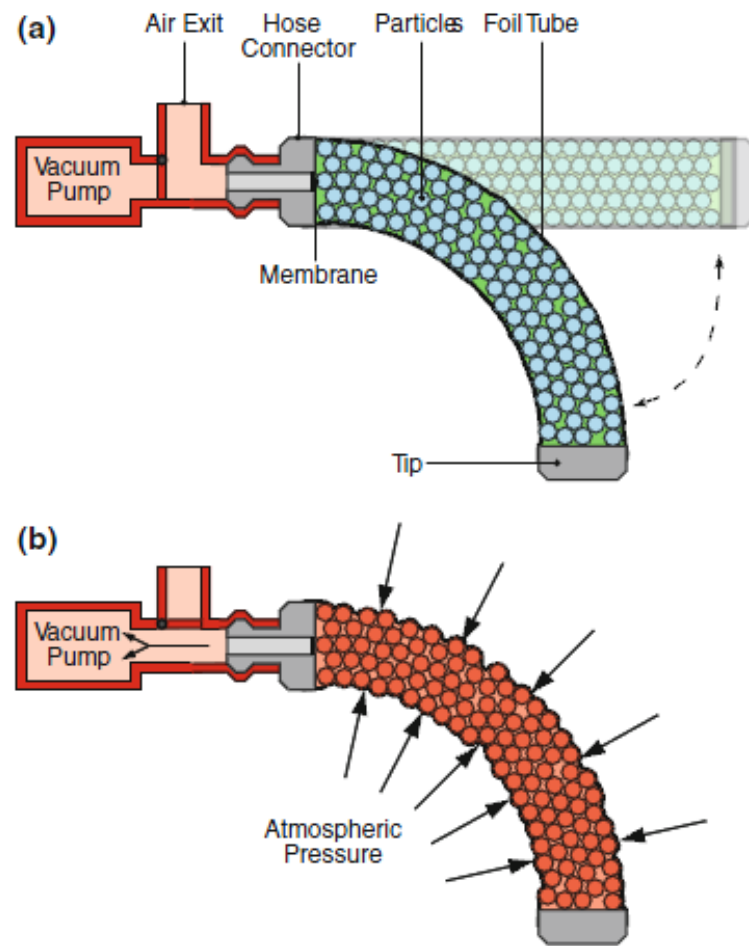
The jamming transition as a general phenomenon is a critical slowing down of the dynamics of a system far from equilibrium due to “overcrowding” of particles. This overcrowding can be affected either by a compression of the particles, or in the case of attractive liquids, by lowering the temperature. With few paths open for particle movement, the dynamics of the system are arrested. It is this **arrest of the dynamics to which jamming refers**. In the case of attractive liquids this phenomenon is better-known as the glass transition, which has been widely studied, but is still poorly understood.



Jamming possibilities

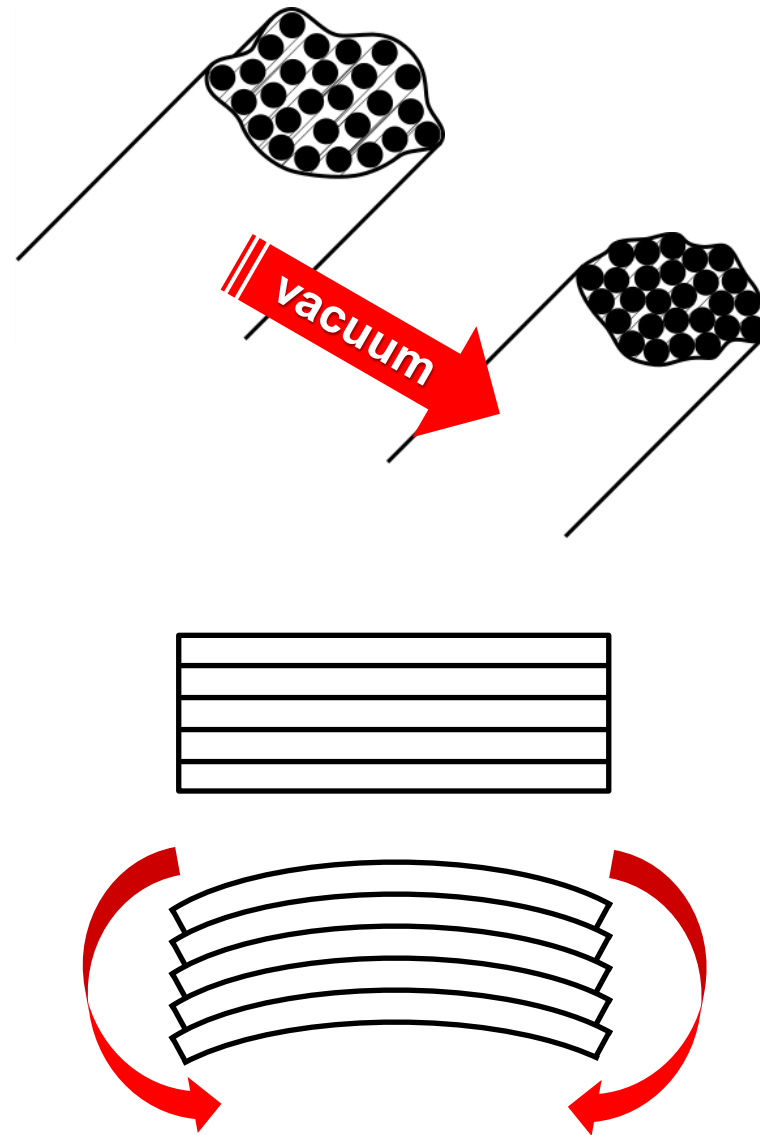
Granular Jamming (GJ)

Granular material + flexible membrane



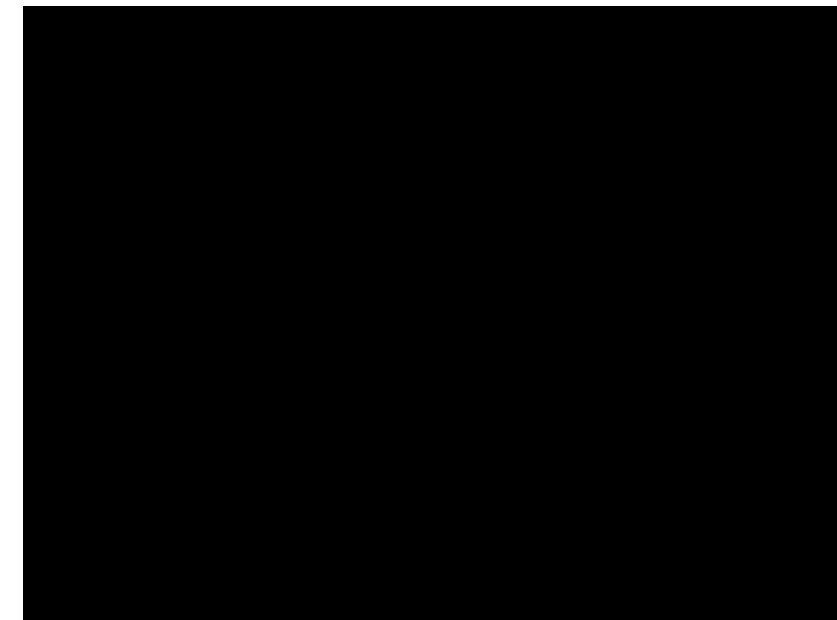
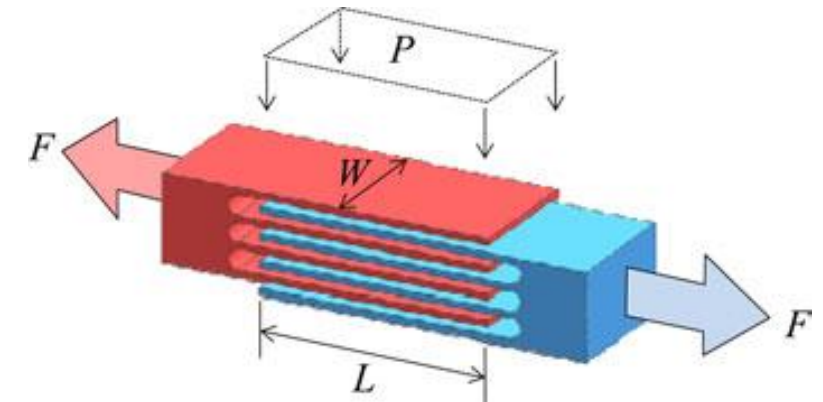
Fibre Jamming (FJ)

Flexible fibres + flexible membrane



Layer Jamming (LJ)

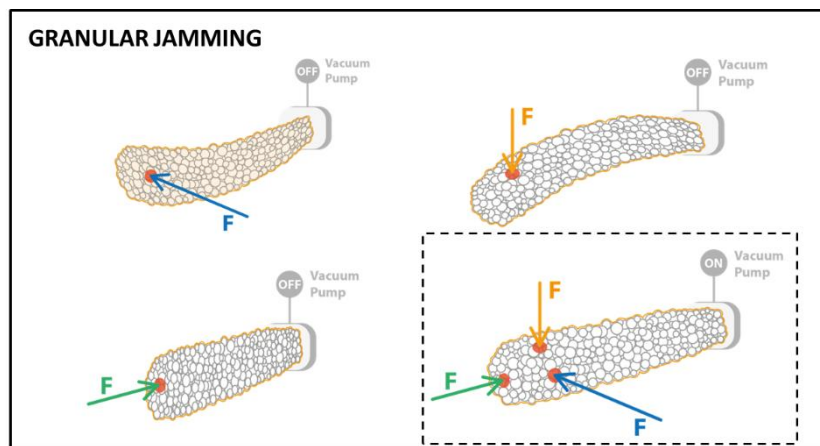
Flexible sheets + flexible membrane



Jamming possibilities

Granular Jamming (GJ)

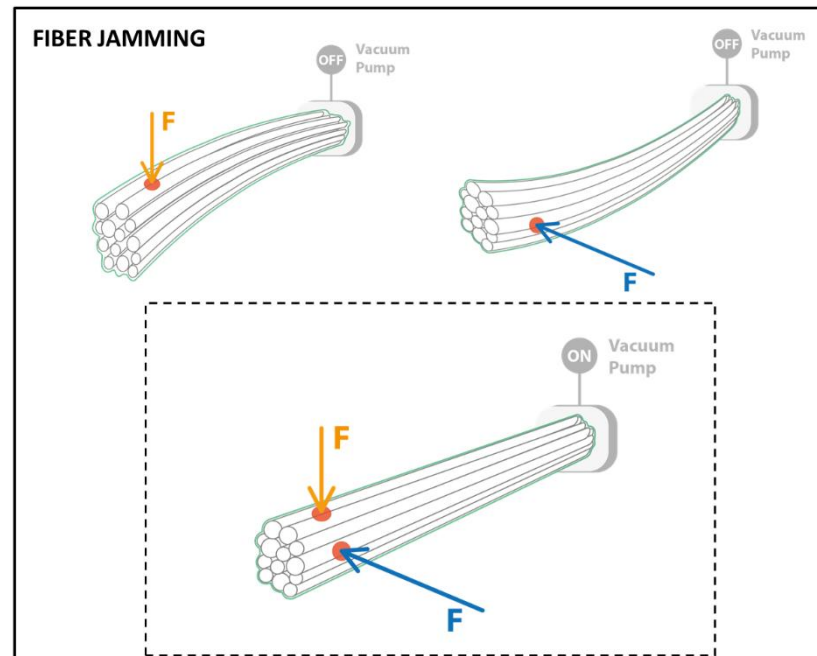
Granular material + flexible membrane



0D $\rightarrow$ 3D

Fibre Jamming (FJ)

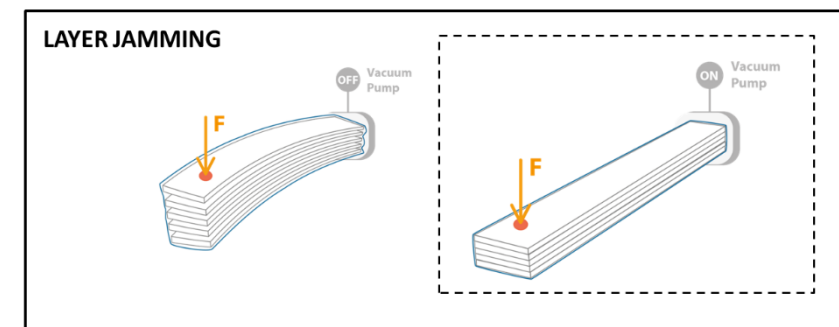
Flexible fibres + flexible membrane



1D $\rightarrow$ 2D

Layer Jamming (LJ)

Flexible sheets + flexible membrane



2D $\rightarrow$ 1D



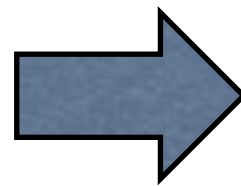
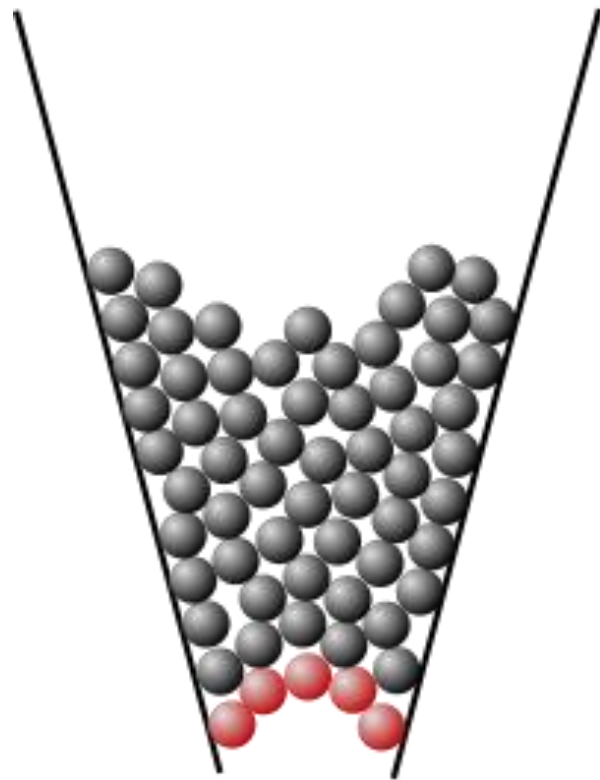
Granular Jamming



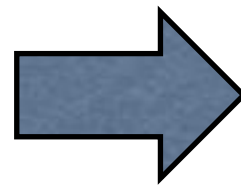
Modeling

“The phenomenon of jamming in granular matter poses fundamental challenges in basic science because **there is no known framework** leading from the microscopic, grain level interactions to the macroscopic properties that reflect collective behavior” *

“the theoretical description of granular systems remains largely a **plethora of different, often contradictory concepts and approaches**” **



SPM (Simple Probabilistic Modeling)



DEM (Distinct Elements Modeling)

* B. Chakraborty and B. Behringer. Jamming of Granular Matter. Encyclopedia of Complexity and Systems Science, pp 4997-5021, 2009.

** A.J. Liu, C.S. O'Hern, L.E. Silbert and S.R. Nagel. Phys. Rev. E, 68:011306, 2003.



Modeling – Statistical mechanics

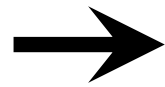
$$\phi \equiv N v_0 / V$$

N particles (grains), each of a fixed volume v_0 , all confined to a box of volume V . A key parameter of the system is the packing fraction, or volume density

$$(1 - \phi)V$$

Free volume

When ϕ is
sufficiently small



$$\Delta P = 0$$

IF external perturbation

THEN bounce around like

particles in a gas or liquid

IF gravitational field

THEN flow like a liquid

**JAMMING
TRANSITION**



When ϕ increases
up to ϕ_J



$$\Delta P > 0$$

particles touch and (except for isolated rattlers) lock into a rigid but disordered structure



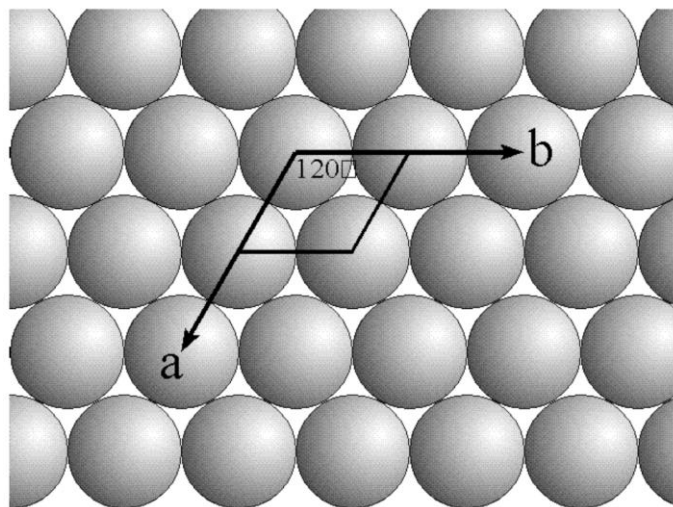
Modeling – Statistical mechanics

Random Close Packing

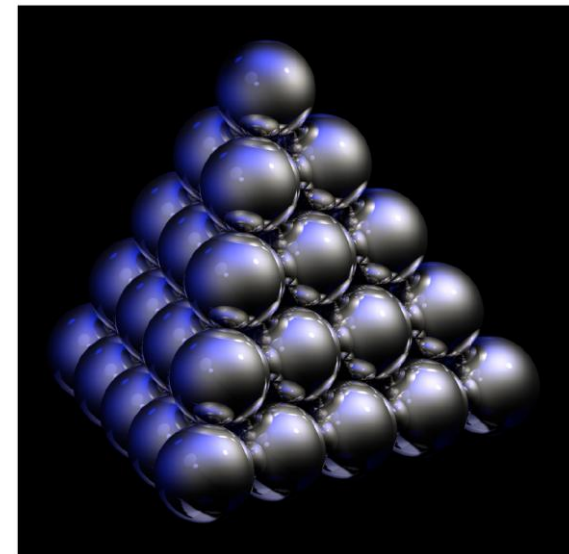
For spherical, frictionless, and rigid (incompressible, non-deformable) particles, the density at which particles jam is often called the ***random close packing*** density ϕ_{RCP}

particles are touching and packed in as tightly as possible

Ordered close packing Bravais lattices are the hexagonal (2D) and face centered cubic - FCC (3D) lattices



$$\phi_{\text{hex}} = \pi / (2\sqrt{3}) = 0.9069$$



$$\phi_{\text{fcc}} = \sqrt{2}\pi / 6 = 0.7405$$



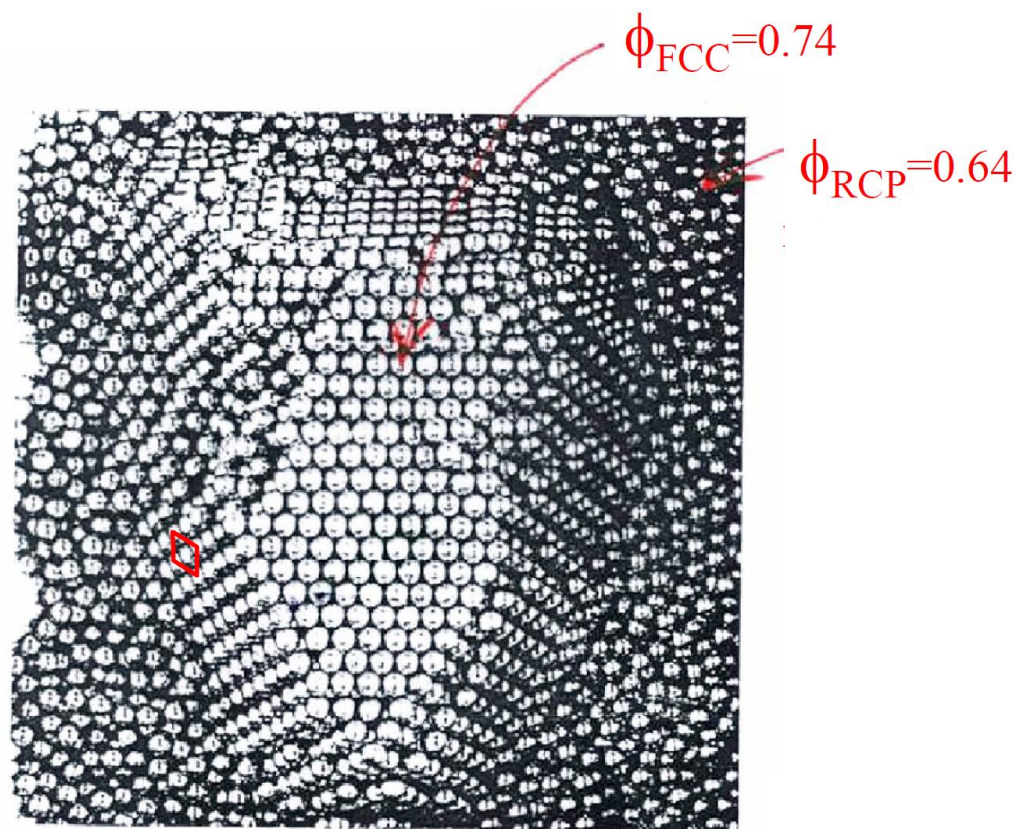
Modeling – Statistical mechanics

Random Close Packing

For spherical, frictionless, and rigid (incompressible, non-deformable) particles, the density at which particles jam is often called the *random close packing* density ϕ_{RCP}

From numerous experiments and numerical simulations

- in 2D $\phi_{\text{RCP}} \simeq 0.84$
- in 3D $\phi_{\text{RCP}} \simeq 0.64$



The randomly closed packed jammed state is therefore in principle only meta-stable. For large N in 3D, however, this meta-stability is extremely stable!

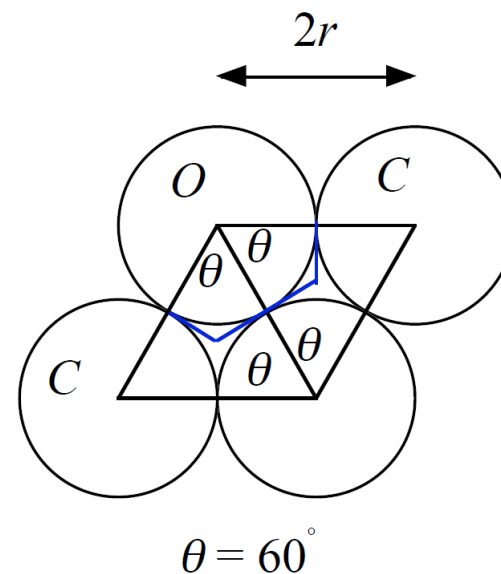
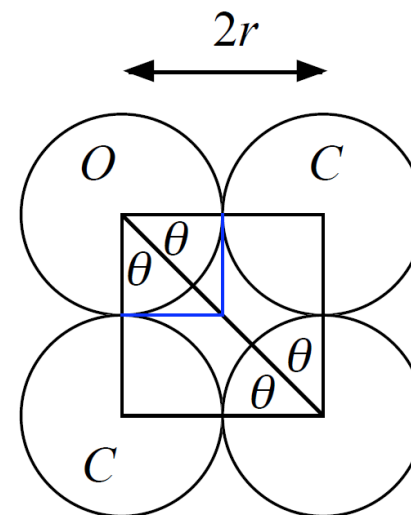
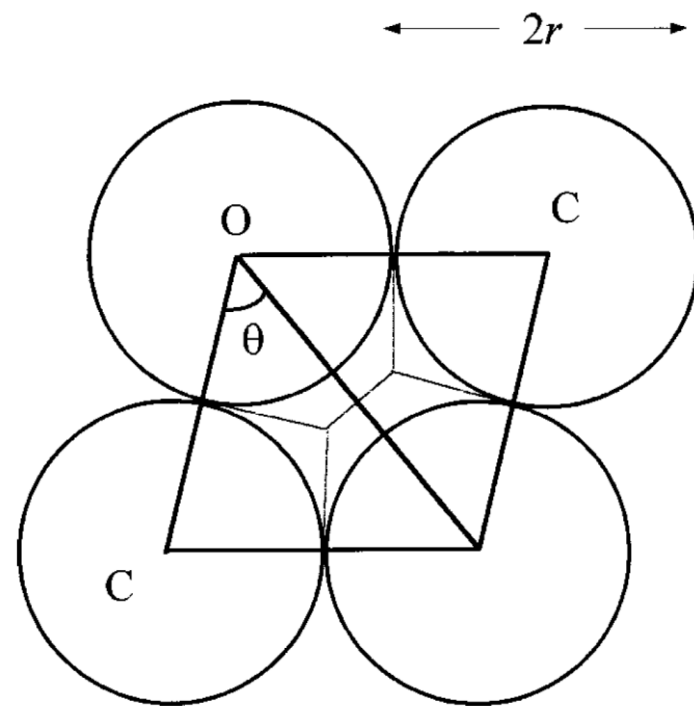
A clear mathematical definition of what is meant by RCP, and a precise analytical calculation of the values of ϕ_{RCP} remains elusive. It remains debated even if there is any such clear mathematical definition.



Modeling – Statistical mechanics

Random Close Packing

A very simple model that gives a good estimate for the value of ϕ_{RCP} in 2D



$$\phi(\theta) = \frac{\frac{1}{2}\pi r^2}{2r^2 \sin 2\theta} = \frac{\pi}{4 \sin 2\theta}$$

$$\theta = \bar{\theta} = 52.5^\circ$$

$$\phi_{RCP} \simeq 0.813$$

By averaging $\phi(\theta)$ instead of evaluating $\phi(\bar{\theta})$

$$\phi_{RCP} \approx \frac{3}{\pi} \int_{\pi/4}^{\pi/3} \frac{\pi}{\sin 2\theta} d\theta = 0.824$$

Extending to 3D: $\phi_{RCP} \approx 0.6394$

In a random close packing, all possible angles $45^\circ \leq \theta \leq 60^\circ$ are equally likely. The average angle is thus $\bar{\theta} = 52.5^\circ$



Modeling – Statistical mechanics

Random Close Packing

Because the RCP state is only metastable, there always exist mechanically stable states of higher density (up to the ordered close packed density). One can always trade off **density vs order**: include a little ordering, to get to a higher density. In an experiment or simulation it is therefore possible that the exact value of ϕ_{RCP} that one finds **may depend somewhat on the protocol one is using to create the jammed state**.

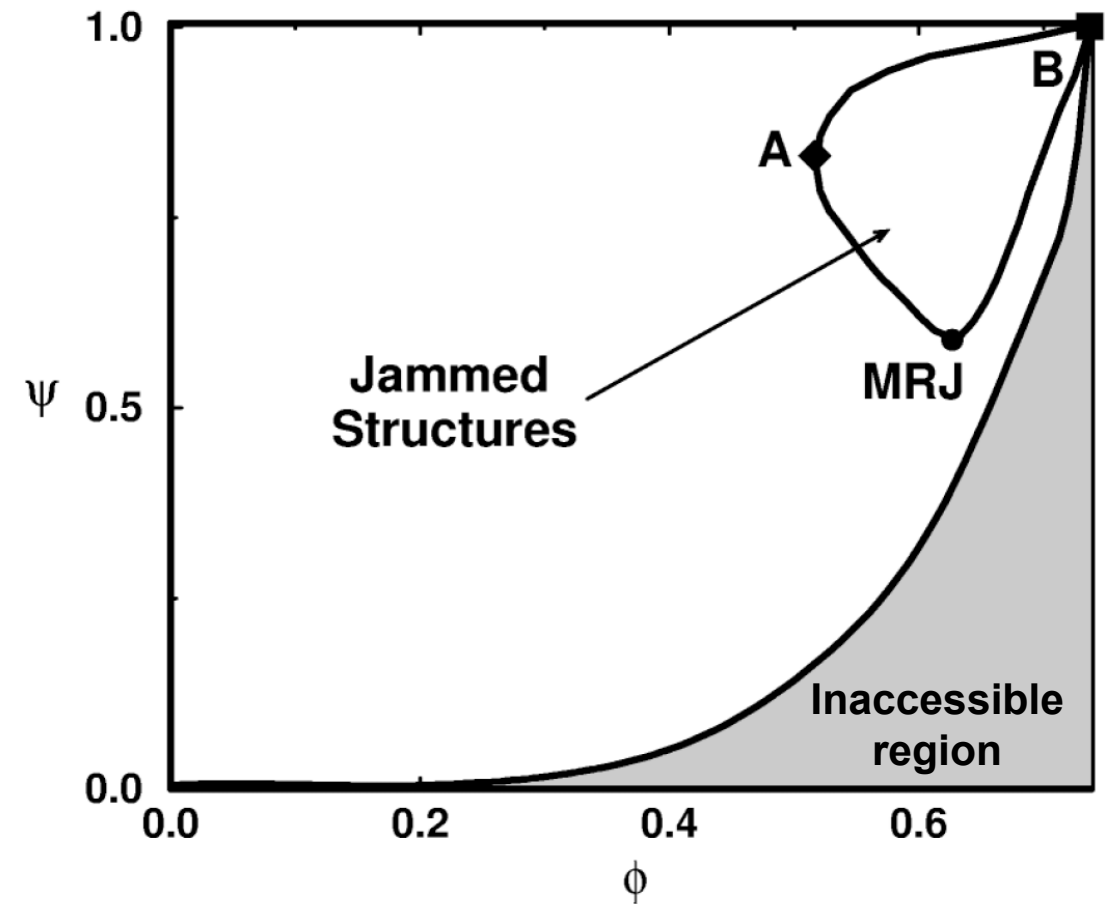
One may question whether the RCP is indeed a mathematically well defined concept...



Modeling – Statistical mechanics

Maximally Random Jammed state

One defines some ordering measure ψ that measures how ordered the packing is ($\psi = 1$ is ordered close packing). One can then draw the region in the $\psi - \phi$ plane where stable jammed packing can occur. The MRJ is defined as the jammed state with the smallest possible value of ψ . Torquato et al.'s numerical simulations find $\phi_{\text{MRJ}} \approx 0.64$, in agreement with accepted values for ϕ_{RCP} .



A: jammed structure with the lowest density

B: the densest ordered jammed structure (e.g., close-packed -fcc- or hexagonal lattice for $d = 3$, depending on the choice for ψ).

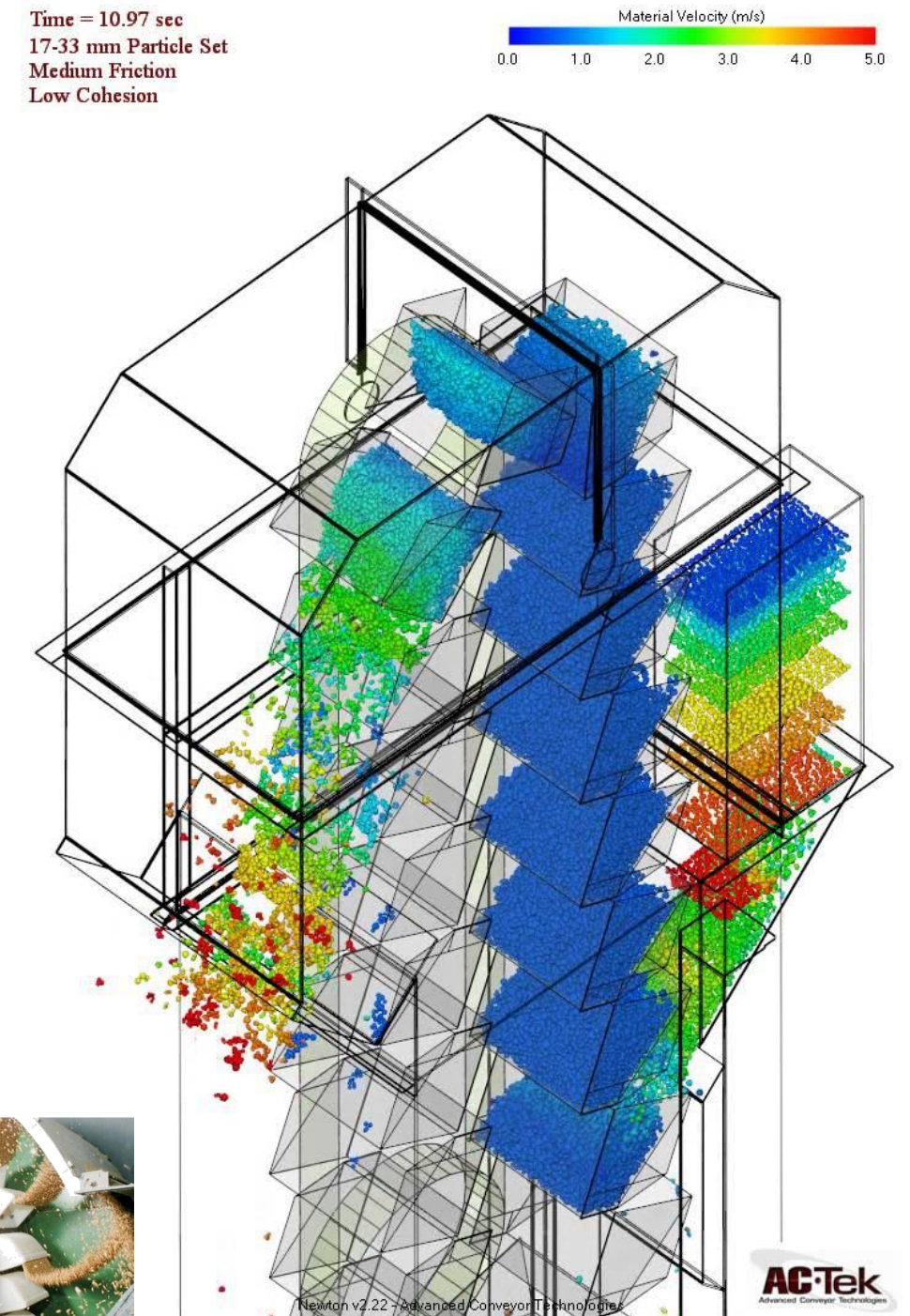
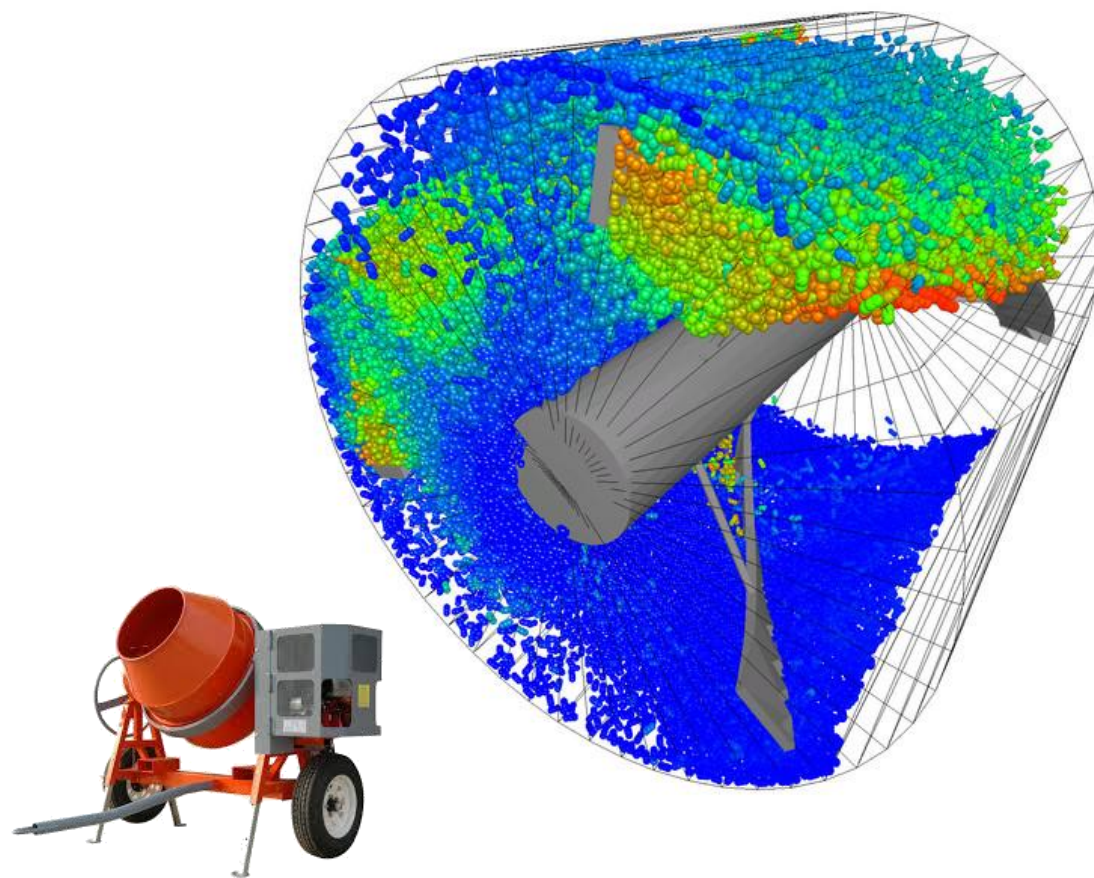
The jammed structure which minimizes the order parameter ψ is the maximally random jammed state.

(Torquato, Truskett, and Debenedetti, PRL 84, 2064 (2000))



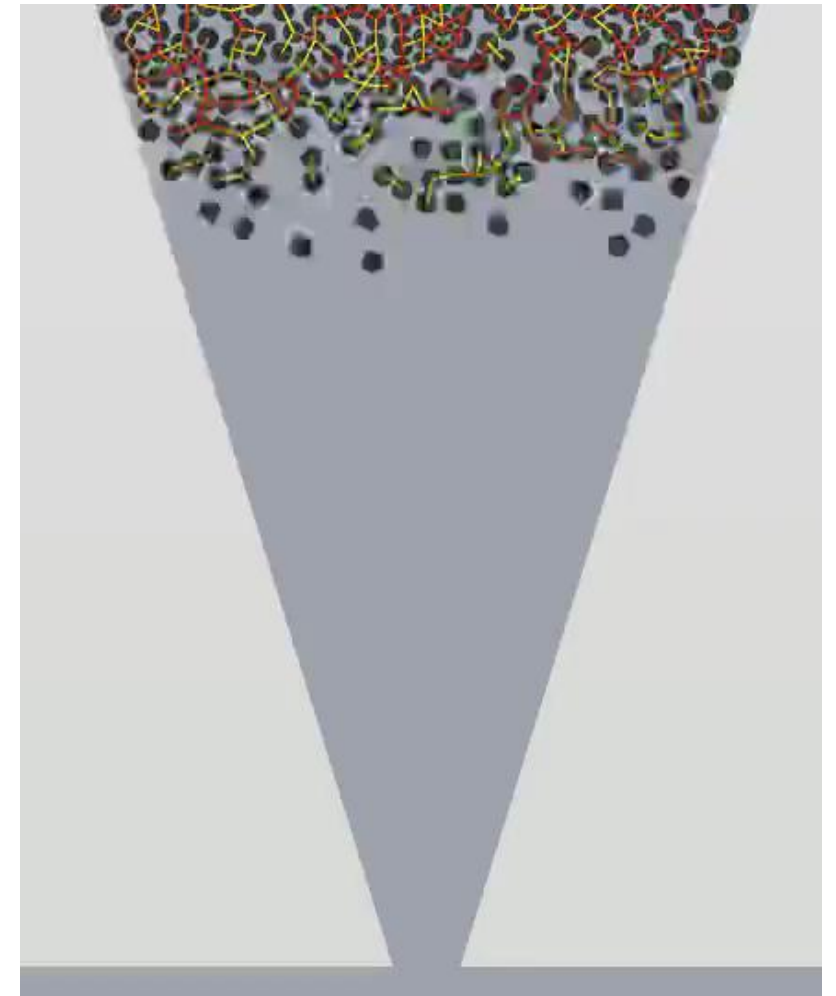
Modeling – Distinct Elements Modeling

Numerical methods for computing the motion and effect of a large number of small particles. Today DEM is becoming widely accepted as an effective method of addressing engineering problems in granular and discontinuous materials, especially in granular flows, powder mechanics, and rock mechanics (geomechanics).



Modeling – Distinct Elements Modeling

1. generating a model
 - a) spatial orientation
 - b) initial velocity
2. forces computed (relevant physical laws and contact models)
3. simulation
 - a) initialization
 - b) explicit time-stepping (nearest neighbor sorting step to reduce the number of possible contact pairs and decrease the computational requirements)
 - c) post-processing



The following forces may have to be considered in macroscopic simulations:

- friction, when two particles touch each other;
- contact plasticity, or recoil, when two particles collide;
- gravity, the force of attraction between particles due to their mass, which is only relevant in astronomical simulations;
- attractive potentials, such as cohesion, adhesion, liquid bridging, electrostatic attraction.



Modeling – Distinct Elements Modeling

Advantages

- DEM can be used to **simulate a wide variety of granular flow and rock mechanics** situations. Several research groups have independently developed simulation software that **agrees well with experimental findings** in a wide range of engineering applications, including adhesive powders, granular flow, and jointed rock masses.
- DEM allows a more **detailed study of the micro-dynamics of powder flows** than is often possible using physical experiments. For example, the force networks formed in a granular media can be visualized using DEM. Such measurements are **nearly impossible in experiments with small and many particles**.

Disadvantages

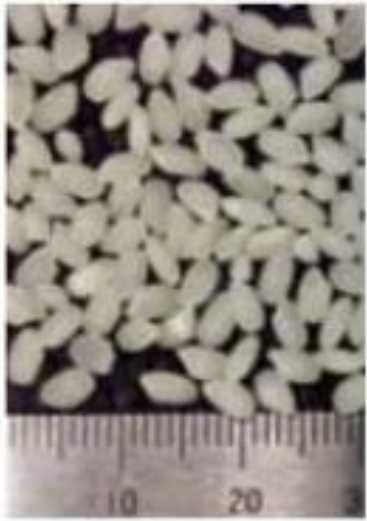
- The maximum number of particles, and duration of a virtual simulation is **limited by computational power**. Typical flows contain billions of particles, but contemporary DEM simulations on large cluster computing resources have only recently been able to approach this scale for sufficiently long time (simulated time, not actual program execution time)



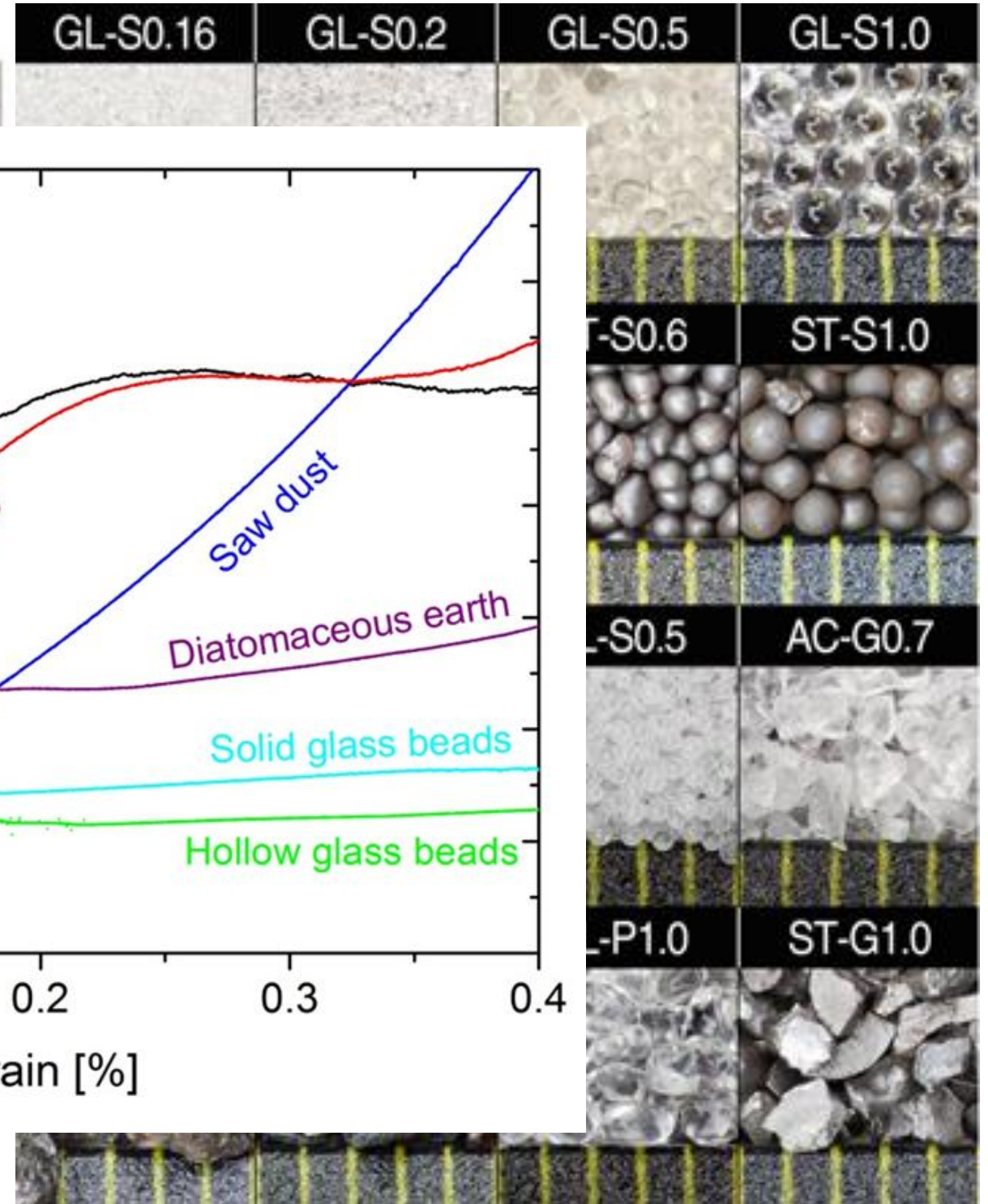
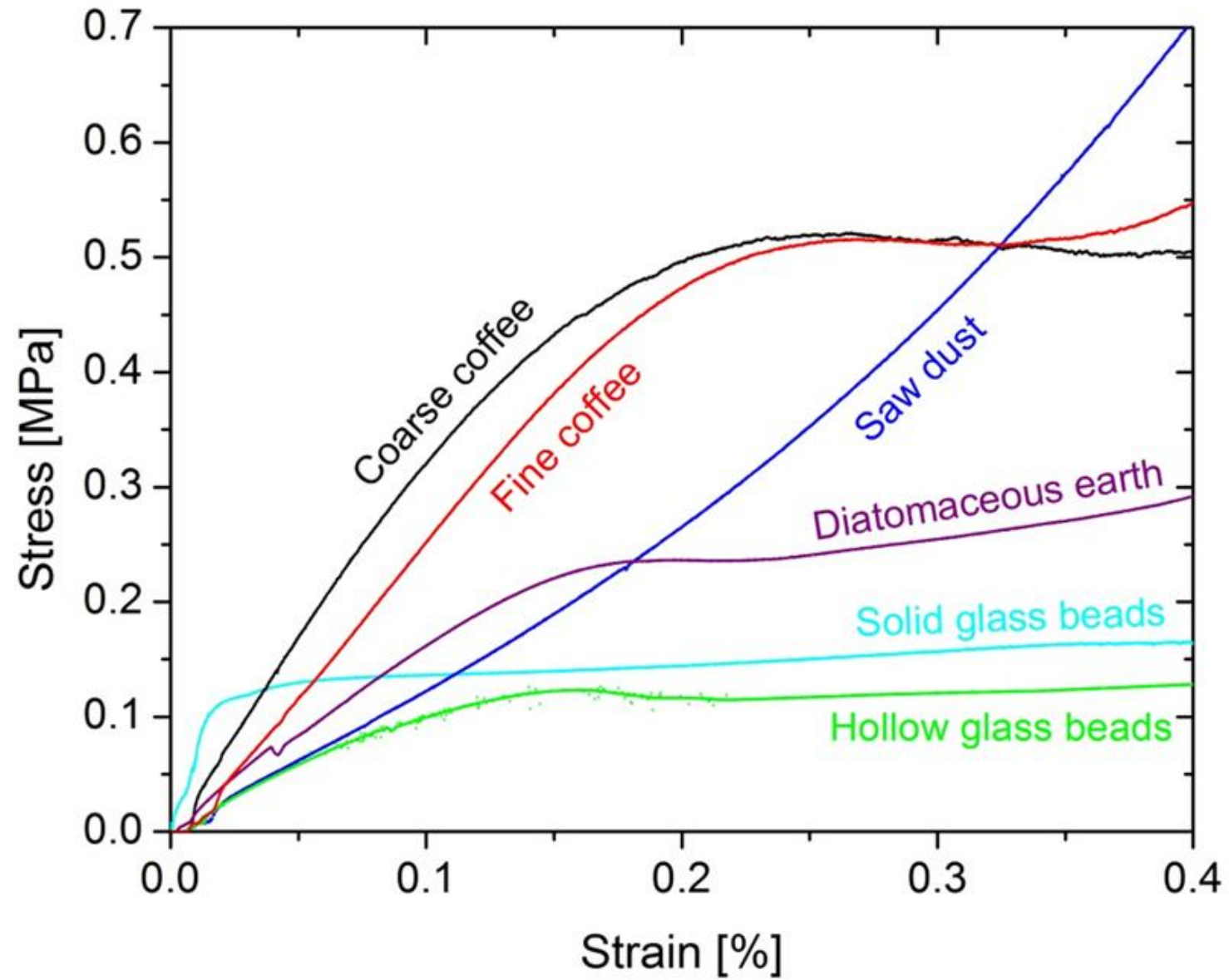
Granular material?



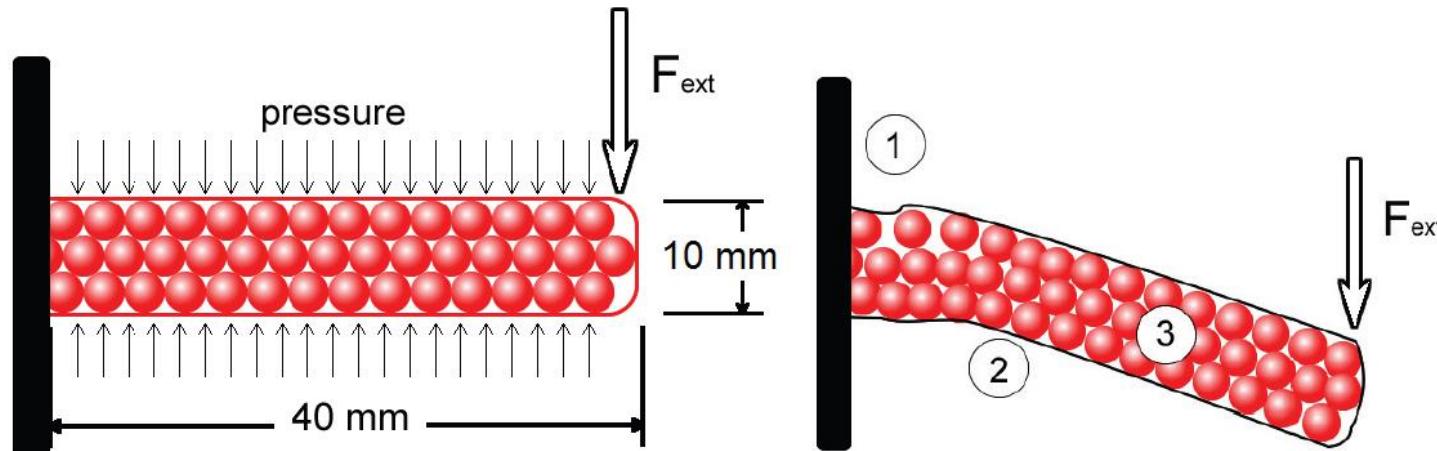
Salt



Rice



Experimental characterization

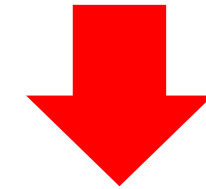


Phenomenologically

Region 1 undergoes tension and granules lose contact with each other

Region 2 experiences compression

Region 3 is where particles remain in their normal configuration.



Fixed-ended solid cylinder
undergoing point load at free tip

$$|M| = |L||F_{ext}|$$

M : total bending moment

L : length of the beam

F_{ext} : externally applied force

$$M = F_{ext}(L - d)$$

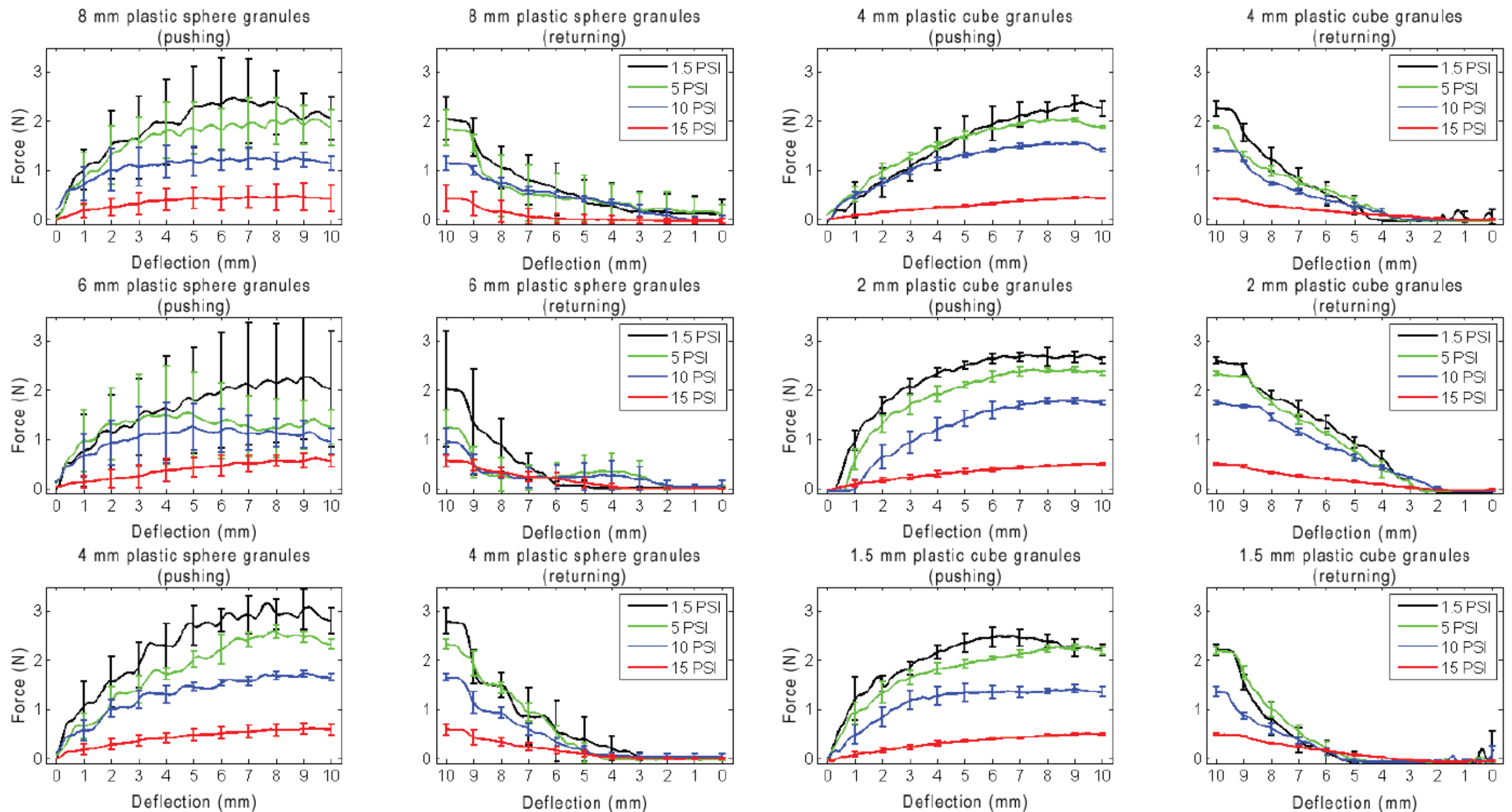
d : distance from the free end

$$y(d) = \frac{F_{ext}d^2}{6EI}(3L - d)$$

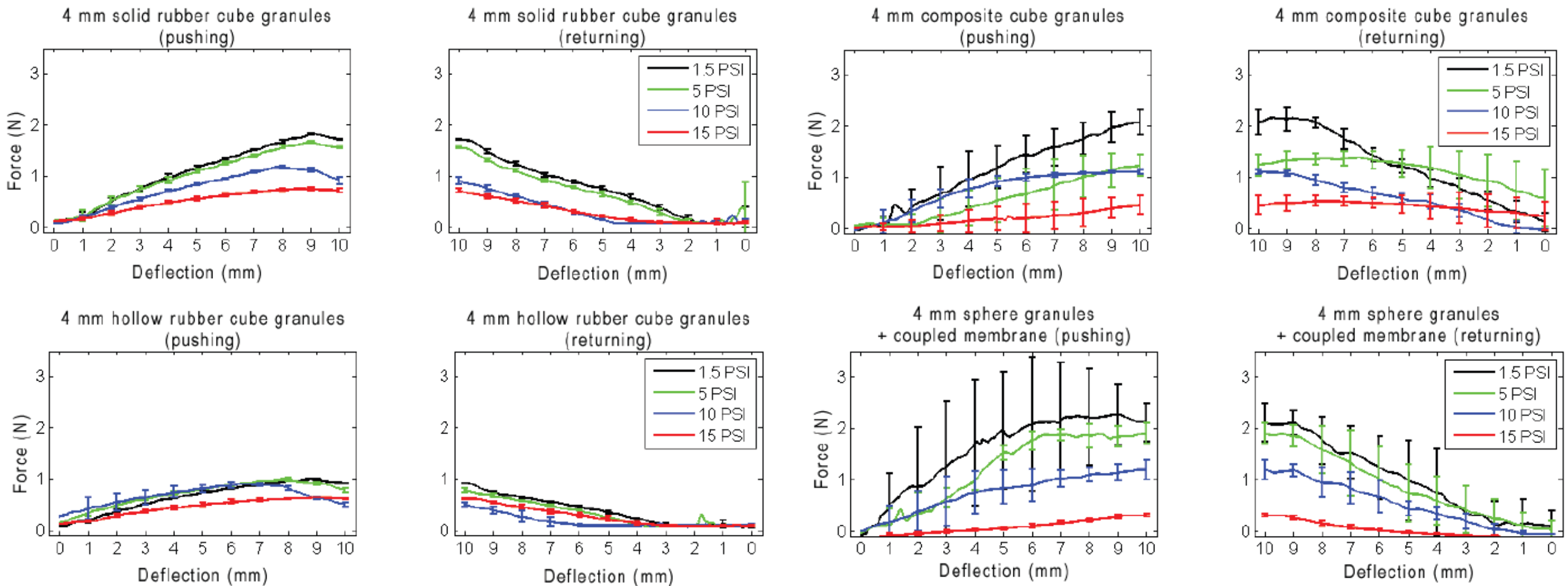
$y(d)$: perpendicular displacement of the
beam along distance d



Experimental characterization



Experimental characterization

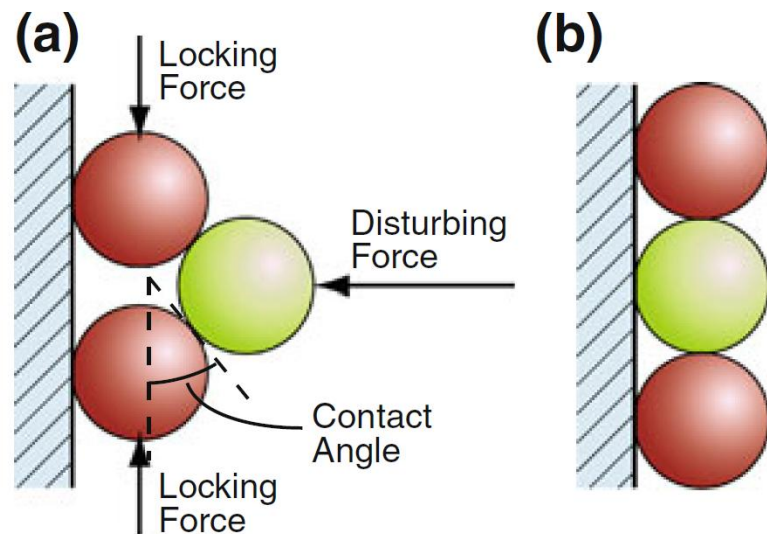


Bead Type	Membrane Type	Force Range (N)	1.5 PSI Hysteresis	1.5 PSI Variability (Total Mean)	# of Trials
8mm Plastic Spheres	Decoupled Latex	0.48 - 2.47	1.25	0.53 (0.32)	10
6mm Plastic Spheres	Decoupled Latex	0.61 - 2.27	1.25	0.57 (0.35)	10
4mm Plastic Spheres	Decoupled Latex	0.61 - 3.16	1.60	0.31 (0.16)	10
4mm Plastic Cubes	Decoupled Latex	0.45 - 2.38	0.94	0.20 (0.09)	5
2mm Plastic Cubes	Decoupled Latex	0.50 - 2.70	1.16	0.10 (0.07)	6
1.5mm Plastic Cubes	Decoupled Latex	0.51 - 2.49	1.65	0.19 (0.11)	6
4mm Solid Rubber Cubes	Decoupled Latex	0.75 - 1.82	0.34	0.05 (0.04)	6
4mm Hollow Rubber Cubes	Decoupled Latex	0.65 - 1.00	0.29	0.02 (0.03)	5
10mm Solid Rubber Block	N/A	5.47 - 5.47	0.30	1.17	8
4mm Composite Cubes	Decoupled Latex	0.53 - 2.16	0.13	0.24 (0.22)	5
4mm Plastic Spheres	Coupled Latex	0.30 - 2.27	0.58	0.67 (0.29)	5

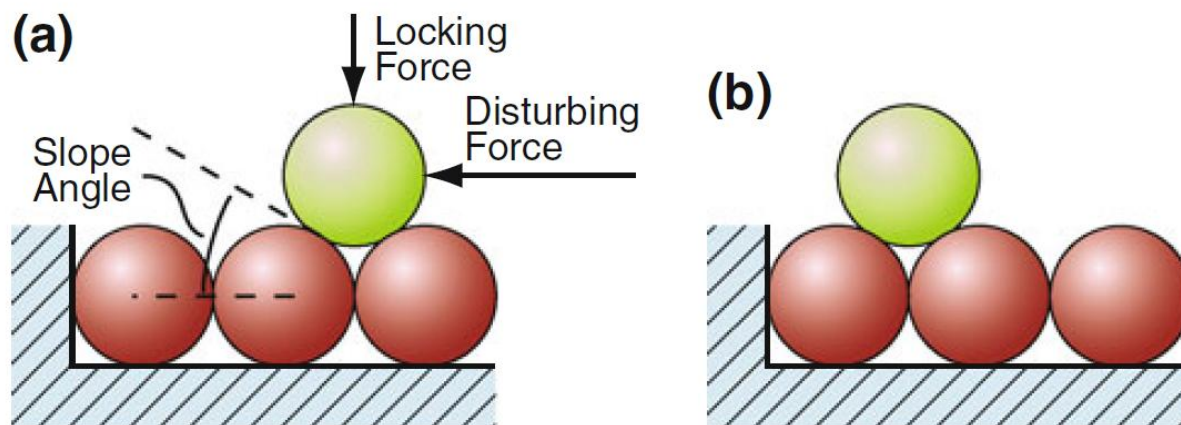


Experimental characterization

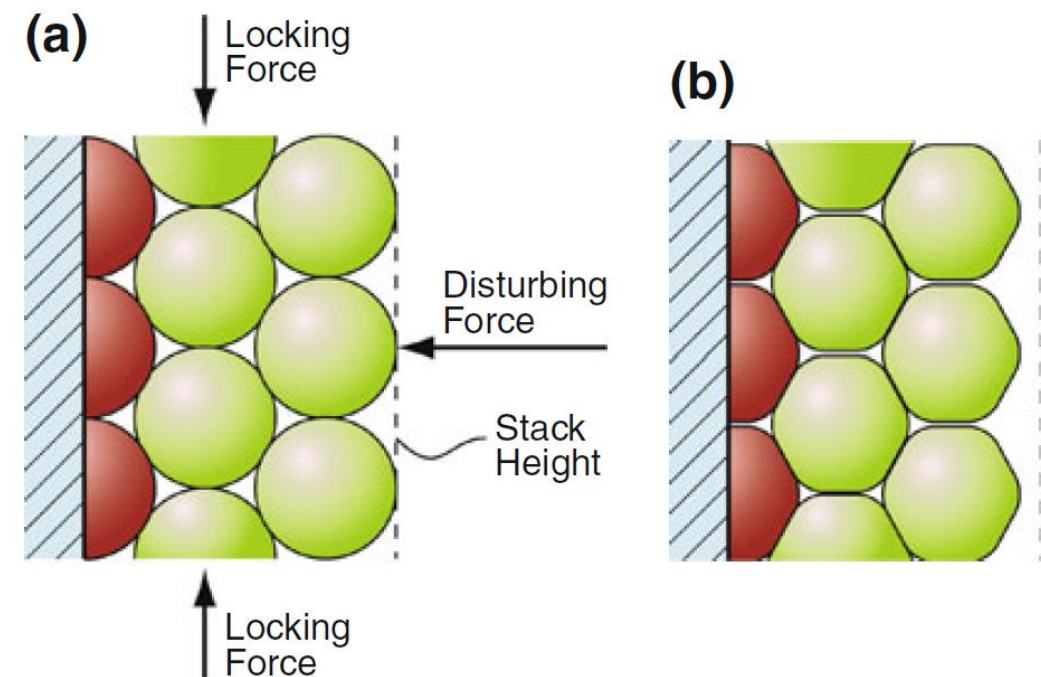
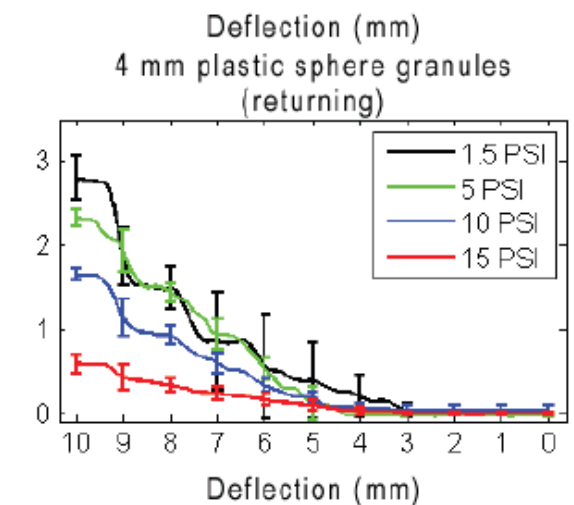
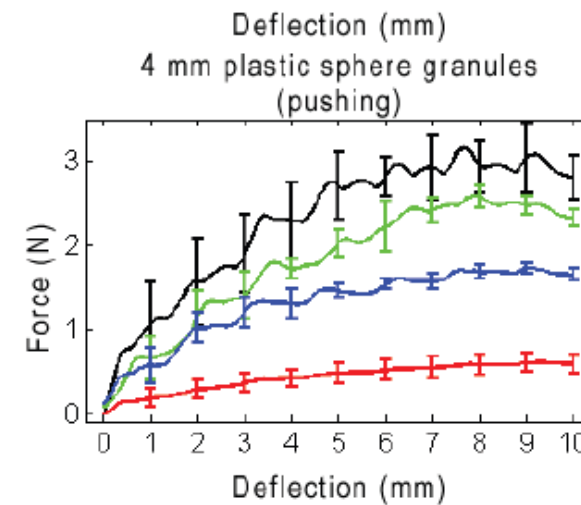
Particle behaviour



Particle intrusion



Particle hopping

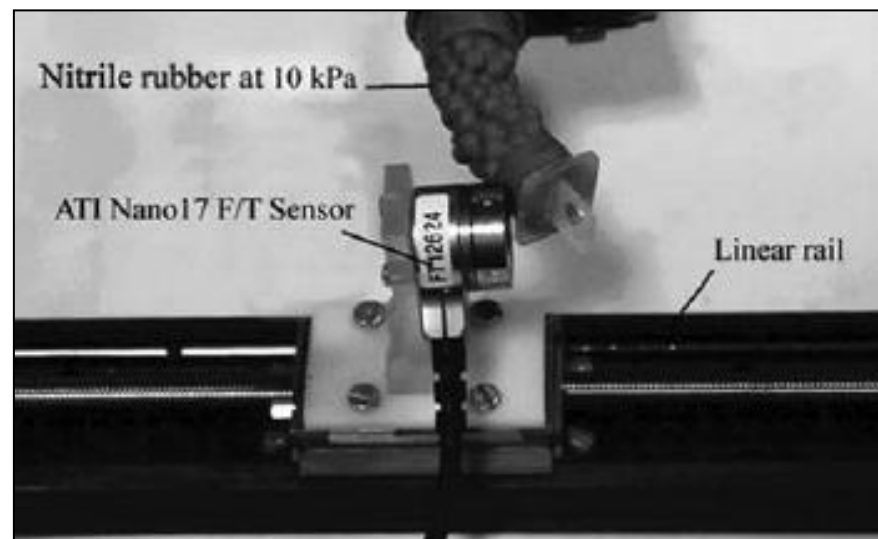
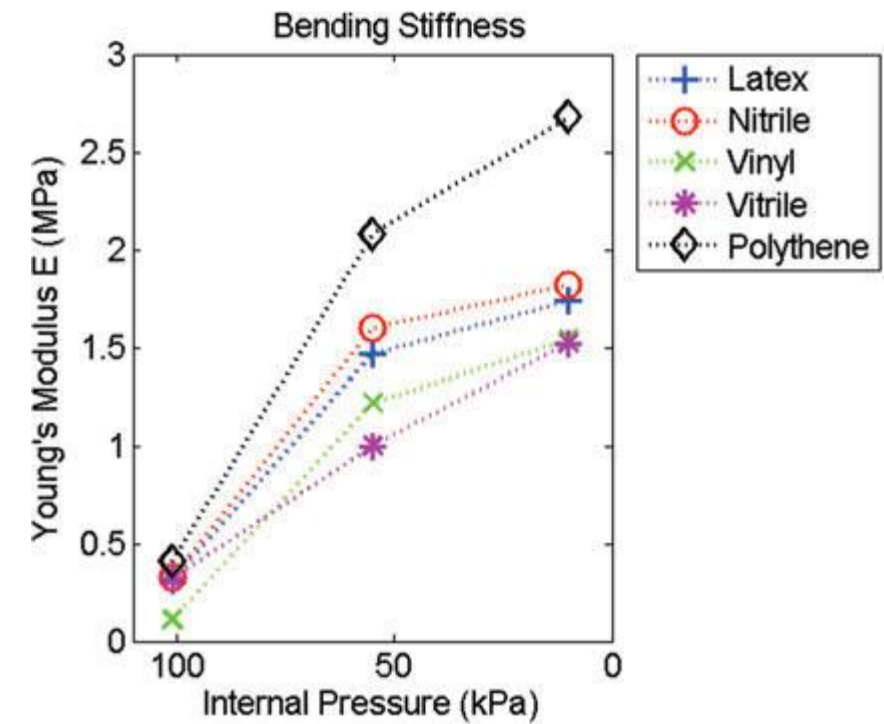
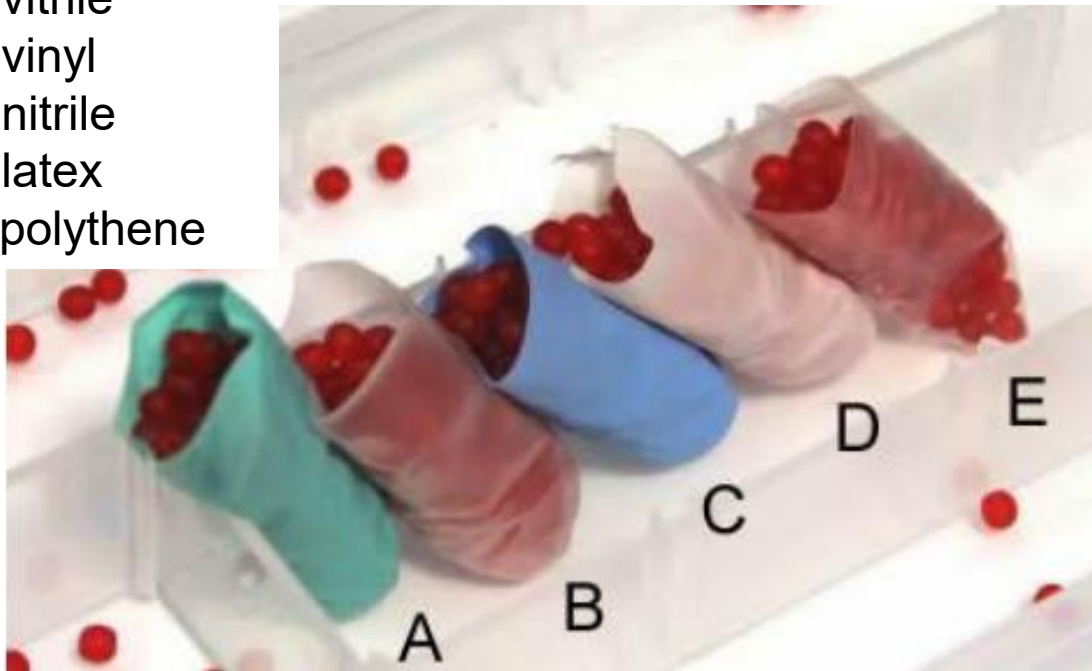


Particle deformation

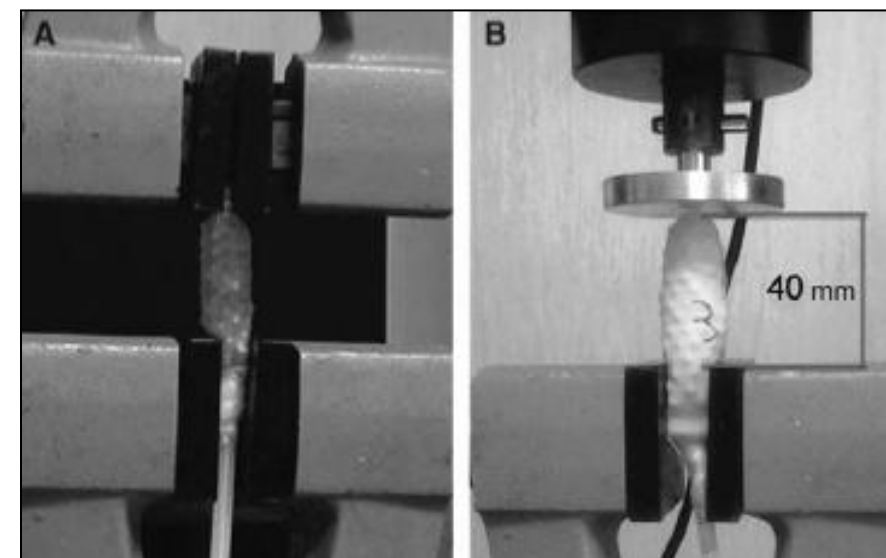


Membrane?

- A vitrile
- B vinyl
- C nitrile
- D latex
- E polythene



Bending test



Tensile test

Compressive test

Jiang A, Ranzani T, Gerboni G, Lekstutyte L, Althoefer K, Dasgupta P, et al. Robotic Granular Jamming: Does the Membrane Matter? Soft Robotics. 2014;1(3):192-201.



Design parameters

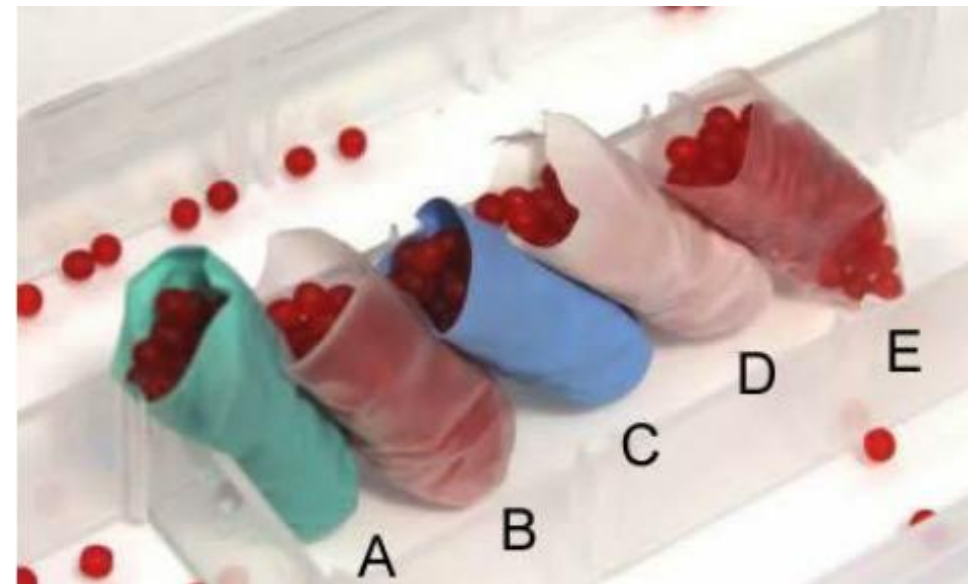
Granular Material

- Size
- Material (deformability)
- Shape
- Texture (friction)



Membrane

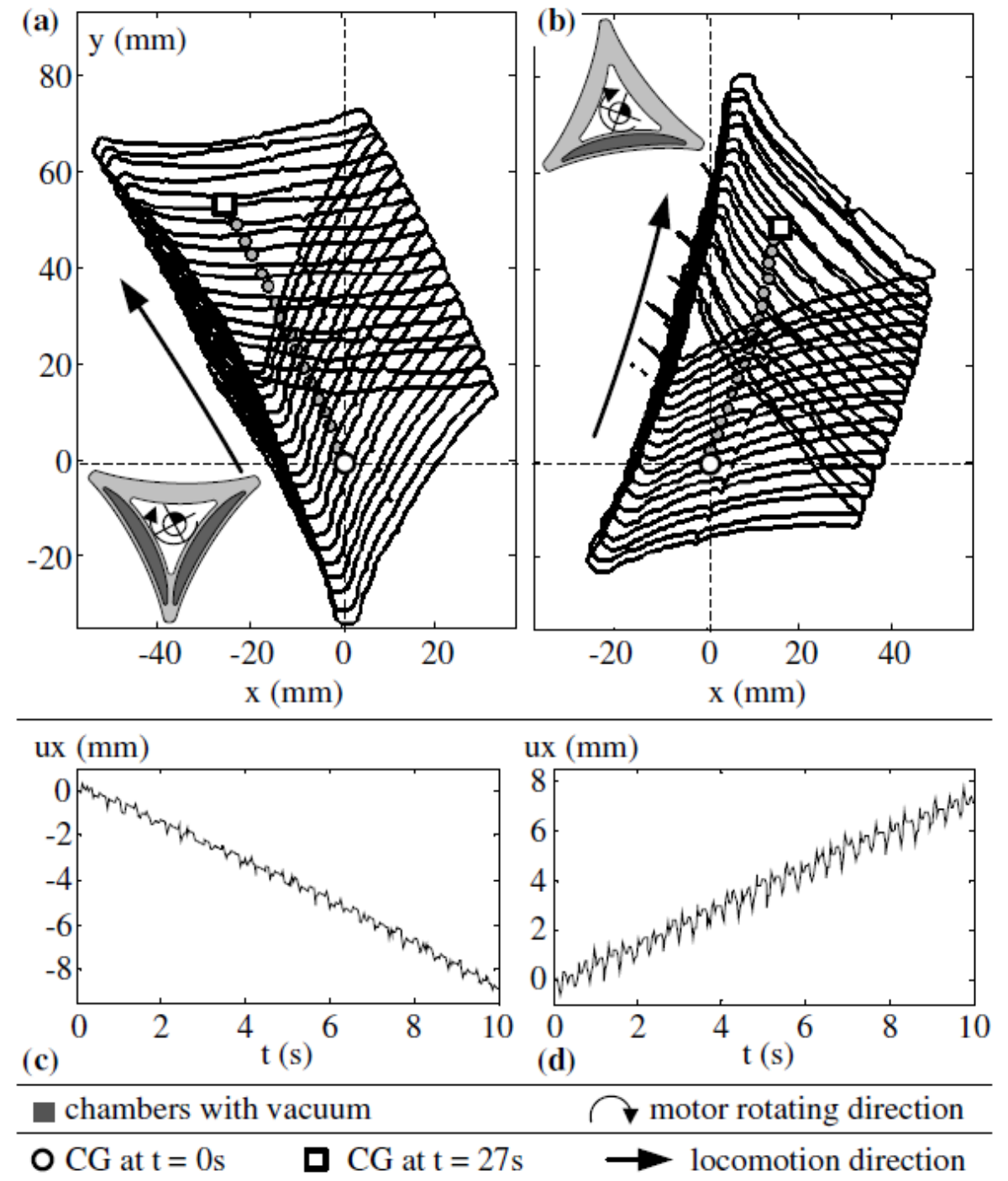
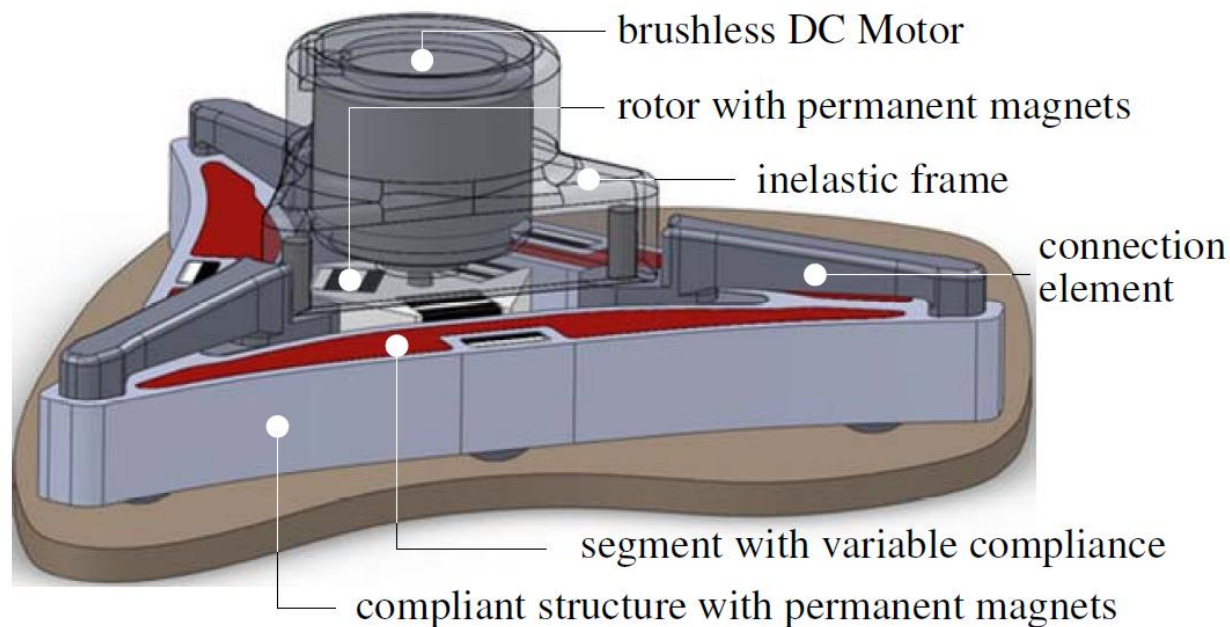
- Thickness
- Material (elasticity)
- Texture (friction)



Applications

Amoeboid

The locomotion of the system is based on the periodic deformation of an elastomeric structure including segments with reversible variable mechanical compliance. The movement direction is defined by an asymmetric configuration of the elastomeric structure induced by segments with differing mechanical compliance.



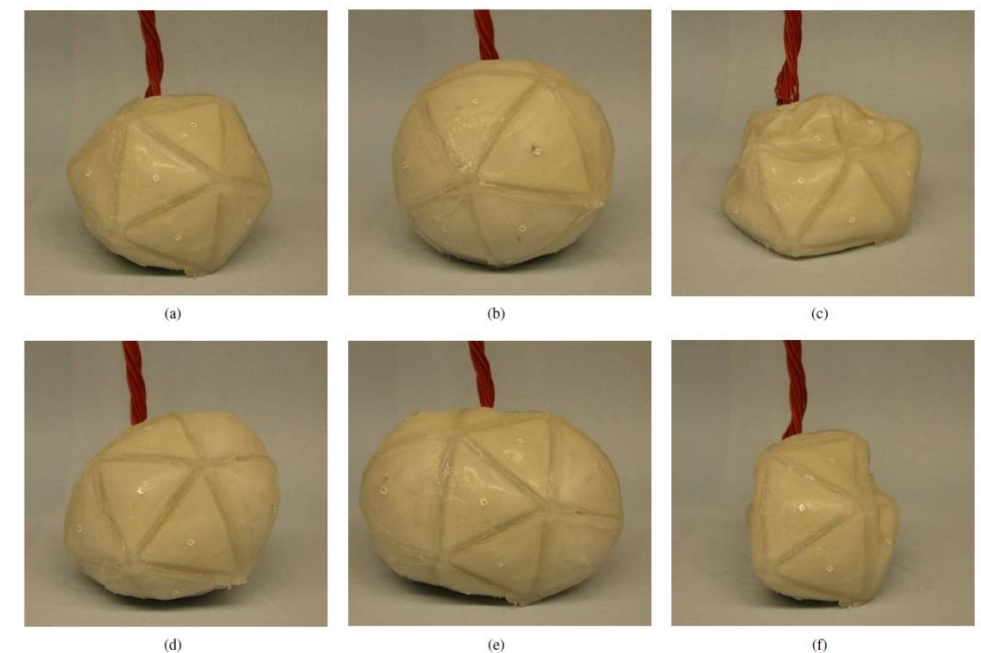
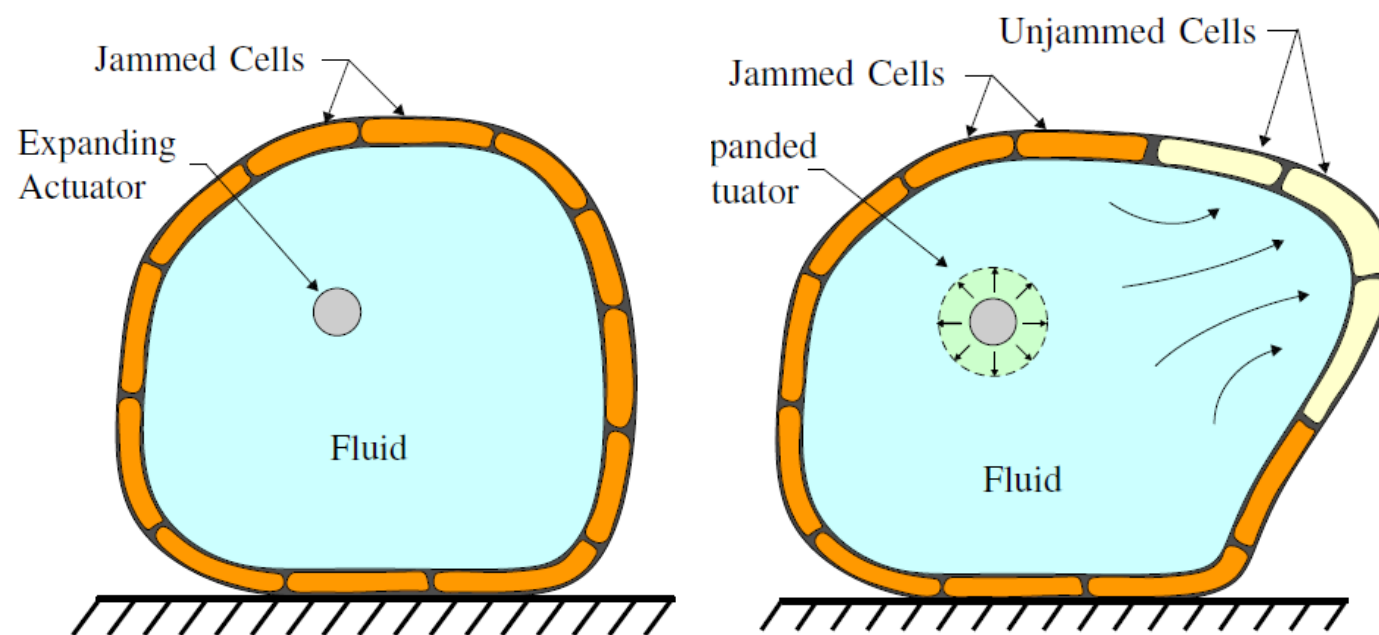
Kaufhold, T.; Bohm, V. & Zimmermann, K. Design of a miniaturized locomotion system with variable mechanical compliance based on amoeboid movement Biomedical Robotics and Biomechatronics (BioRob), 2012 4th IEEE RAS EMBS International Conference on, 2012, 1060-1065



Applications - GJ

JamBot

Soft robot that utilizes particulate jamming technology to modulate the direction and magnitude of the work performed by a single central actuator. Jamming “activators” modulate work by jamming and unjamming a granular medium coupled to a core actuator.



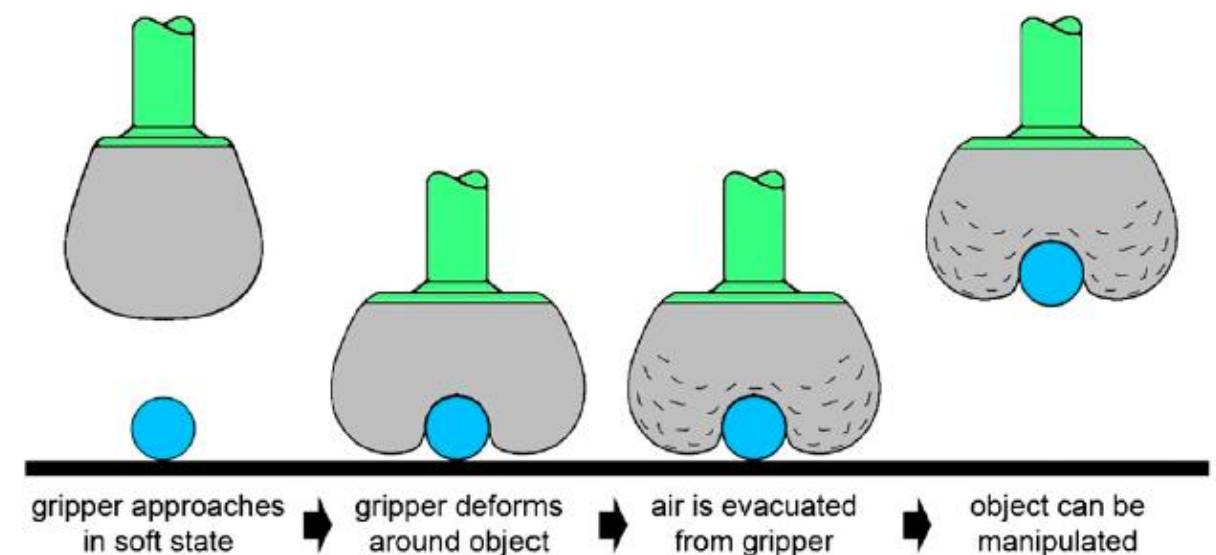
E. Steltz, A. Mozeika, N. Rodenberg, E. Brown, and H. M. Jaeger, "JSEL: Jamming Skin Enabled Locomotion," in Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 10-15 Oct. 2009; pp. 5672-5677 (2009)



Applications - GJ

Universal gripper

A single mass of granular material, pressed onto a target object, flows around it and conforms to its shape. Upon application of a vacuum the granular material contracts and hardens quickly to pinch and hold the object without requiring sensory feedback.



Brown, E.; Rodenberg, N.; Amend, J.; Mozeika, A.; Steltz, E.; Zakin, M. R.; Lipson, H. & Jaeger, H. M. Universal robotic gripper based on the jamming of granular material Proceedings of the National Academy of Sciences, 2010, 107, 18809-18814

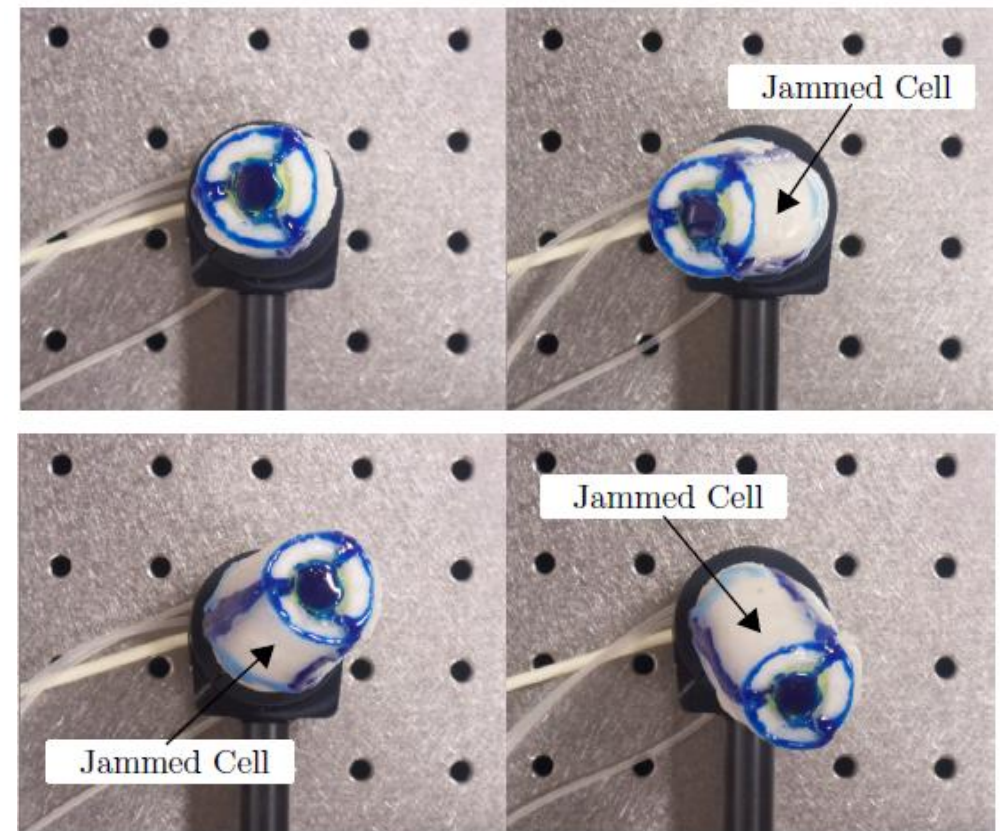
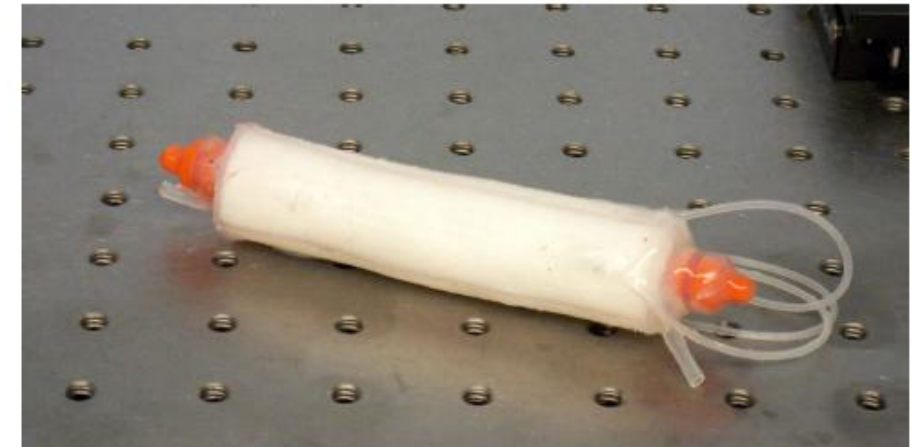
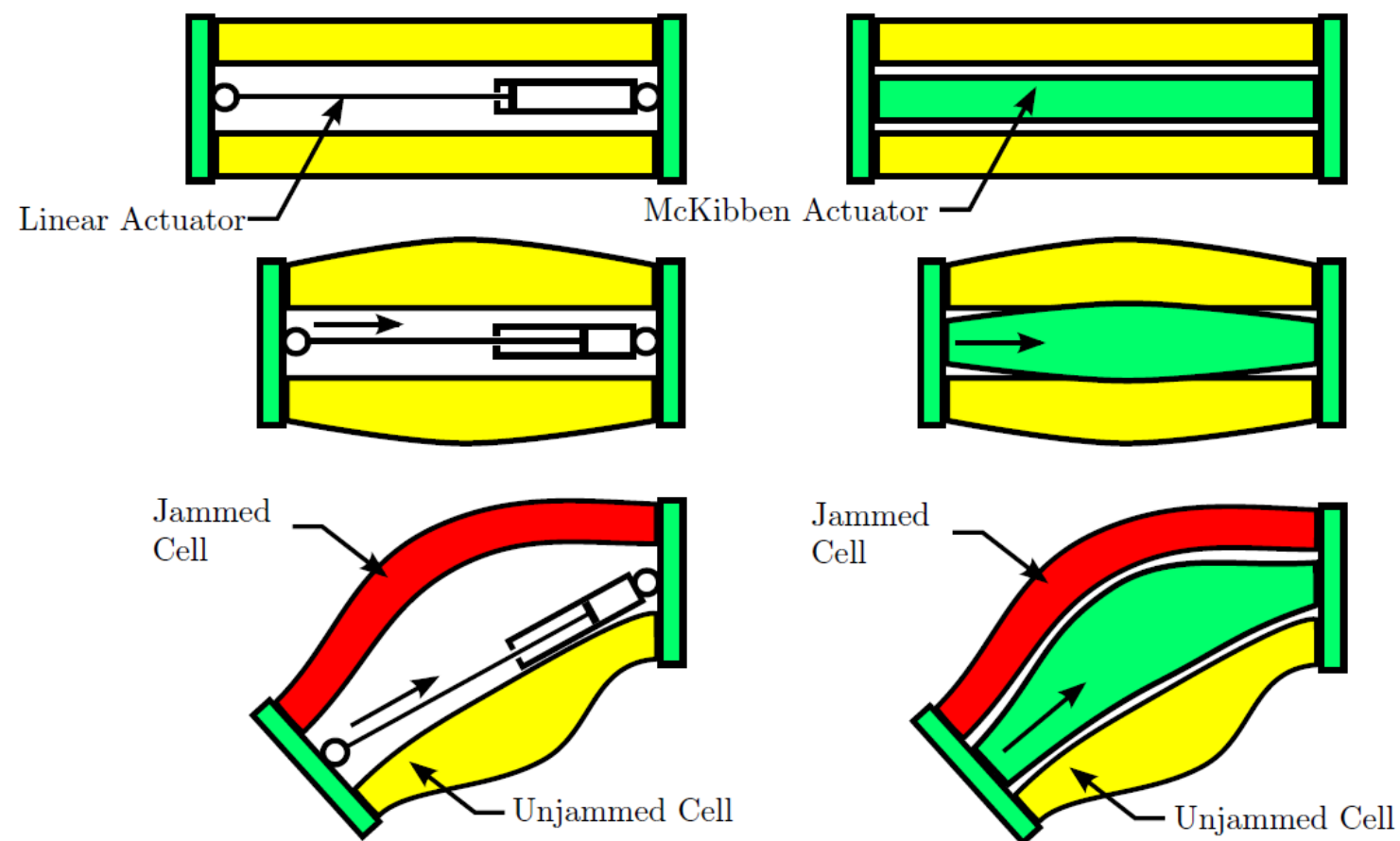
Amend, J.; Brown, E.; Rodenberg, N.; Jaeger, H. & Lipson, H. A Positive Pressure Universal Gripper Based on the Jamming of Granular Material Robotics, IEEE Transactions on, 2012, 28, 341-350



Applications - GJ

JMU

Jamming Modulated Unimorph (JMU) uses a single linear actuator and a discrete number of jamming cells to turn the 1 degree of freedom (DOF) linear actuator into a multi DOF bending actuator.



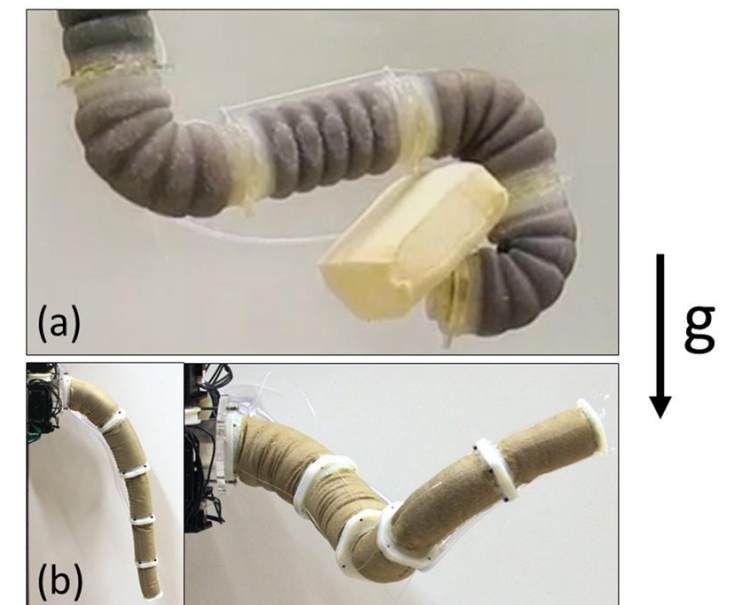
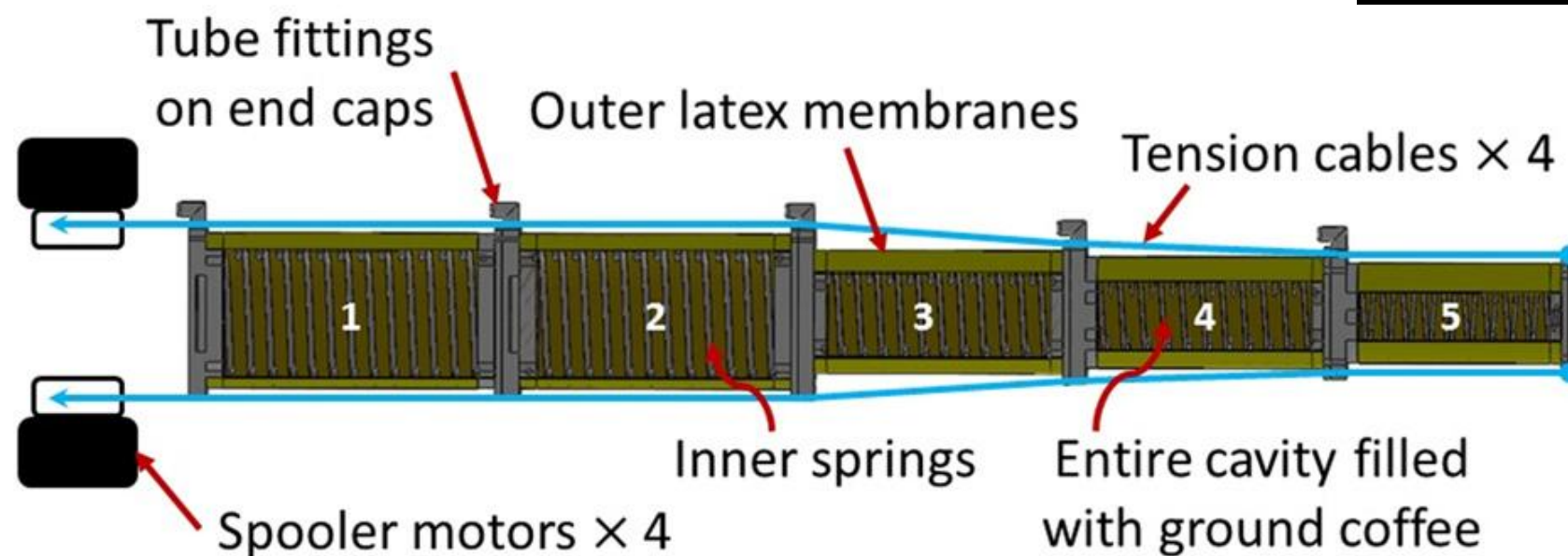
Steltz, E.; Mozeika, A.; Rembisz, J.; Corson, N. & Jaeger, H. M. Jamming as an enabling technology for soft robotics Proc. SPIE, Electroactive Polymer Actuators and Devices (EAPAD) 2010, 2010, 7642, 764225-764225-9



Applications - GJ

MIT manipulator

Local tunable stiffness is coupled with off-board spooler motors and tension cables to achieve complex manipulator configurations. Tunable stiffness is achieved by reversible jamming of granular media, which causes the grains to transition between solid-like states and liquid-like ones.



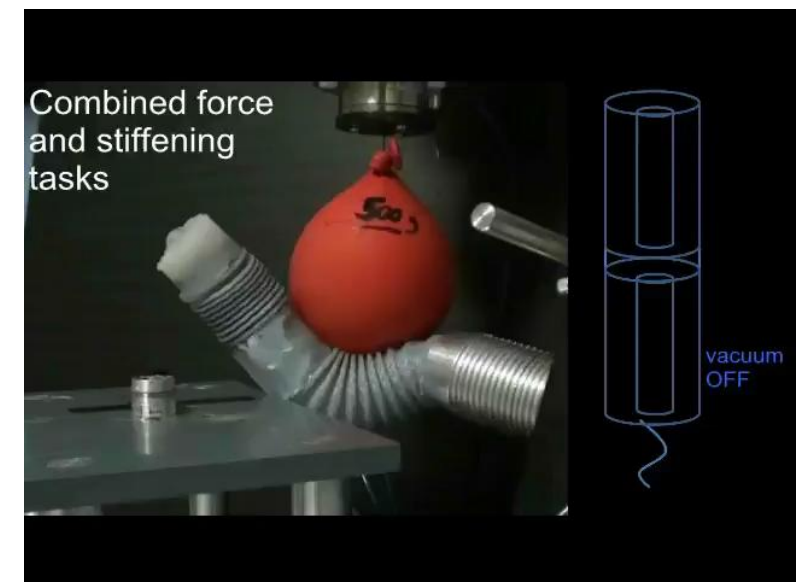
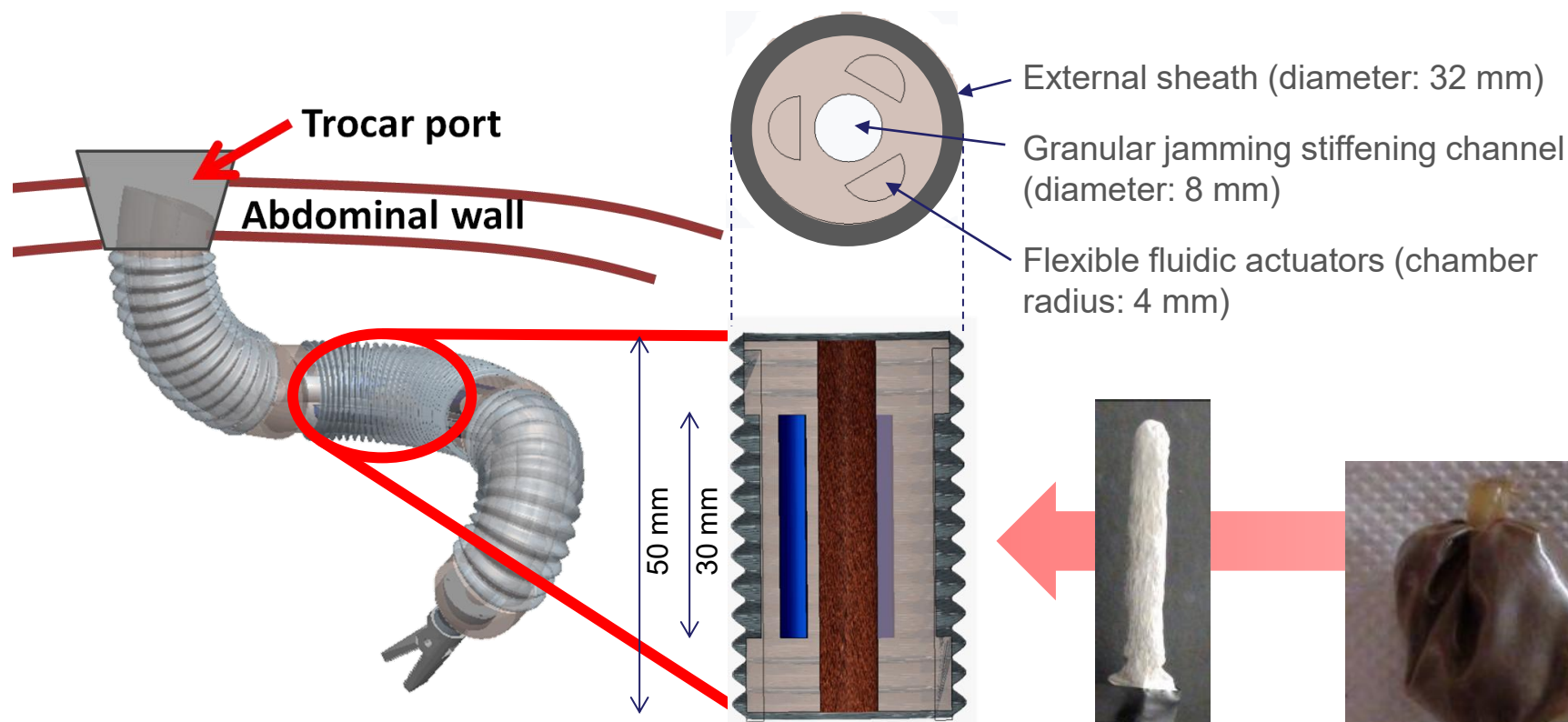
Cheng, N.; Lobovsky, M.; Keating, S.; Setapen, A.; Gero, K.; Hosoi, A. & Iagnemma, K. Design and Analysis of a Robust, Low-cost, Highly Articulated manipulator enabled by jamming of granular media Robotics and Automation (ICRA), 2012 IEEE International Conference on, 2012, 4328-4333



Applications - GJ

STIFF-FLOP

The proposed manipulator is based on a series of identical modules, each one consisting of a silicone tube with pneumatic chambers for allowing 3D motion and one central channel for the implementation of the granular jamming phenomenon for stiffening. The silicone is covered by a novel bellows-shaped braided structure maximizing the bending still limiting lateral expansion.



coffee as granular material and latex membrane

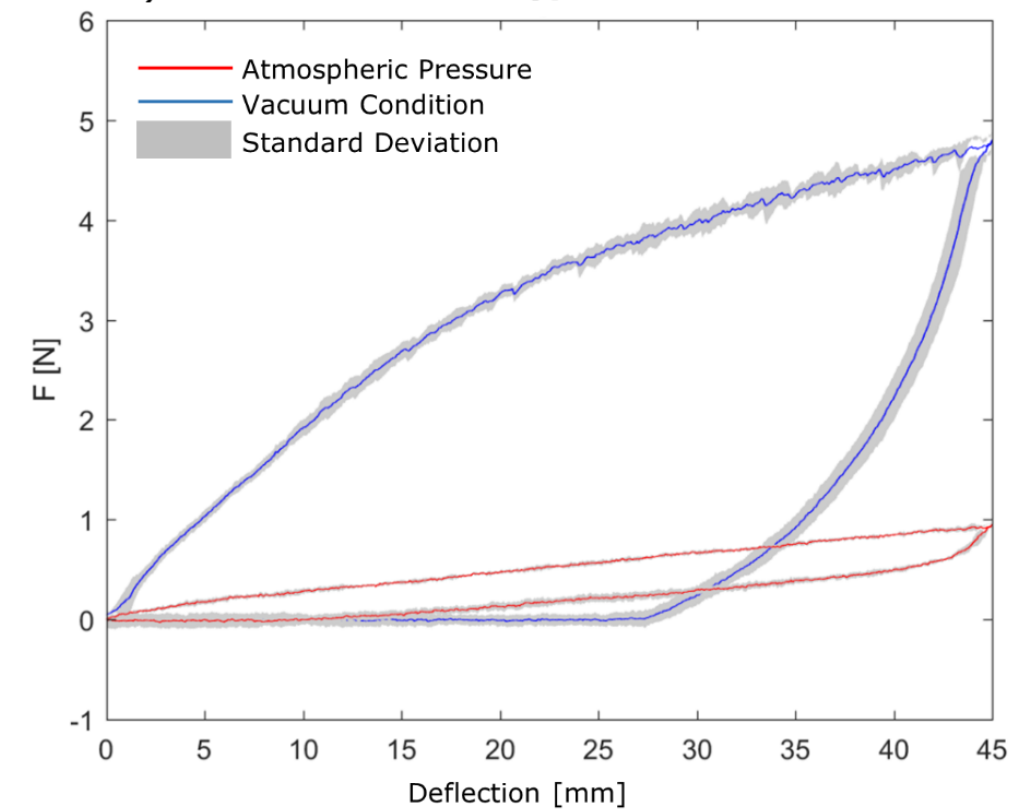
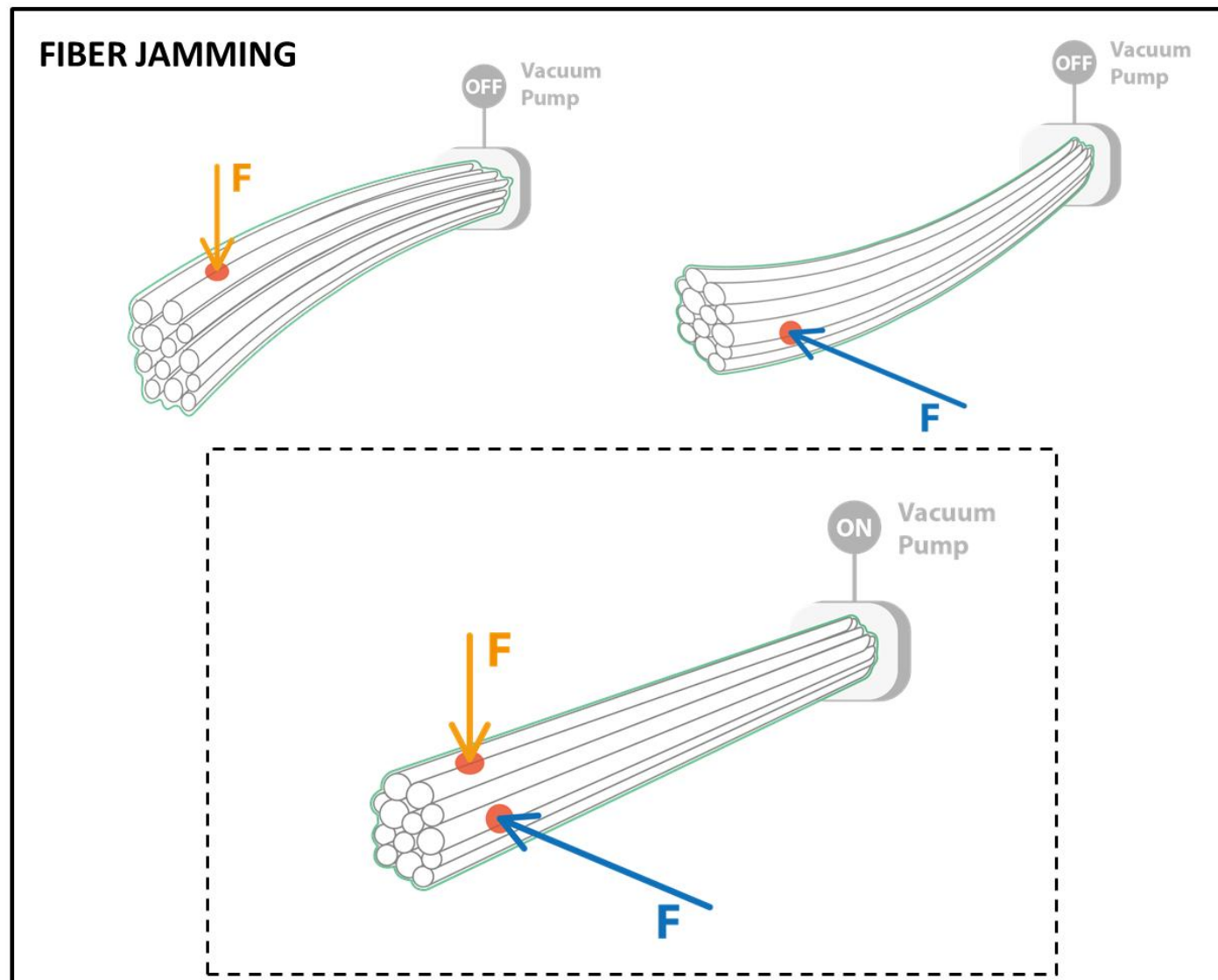
Cianchetti, M.; Ranzani, T.; Gerboni, G.; Nanayakkara, T.; Althoefer, K.; Dasgupta, P. & Menciassi, A. Soft robotics technologies to address shortcomings in today's minimally invasive surgery: the stiff-flop approach Soft Robotics, Mary Ann Liebert, Inc. 140 Huguenot Street, 3rd Floor New Rochelle, NY 10801 USA, 2014, 1, 122-131



Fiber Jamming



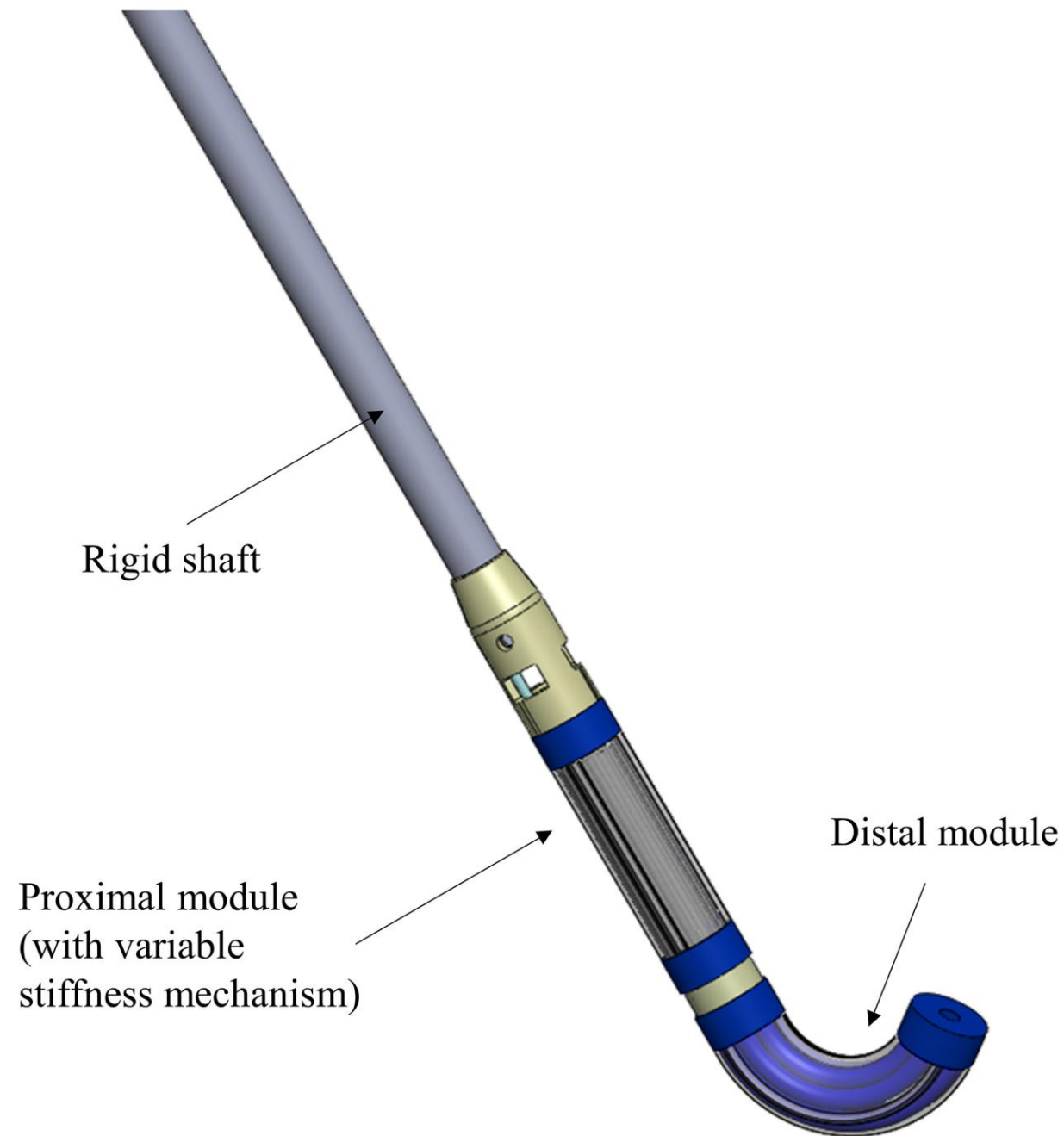
Fiber jamming



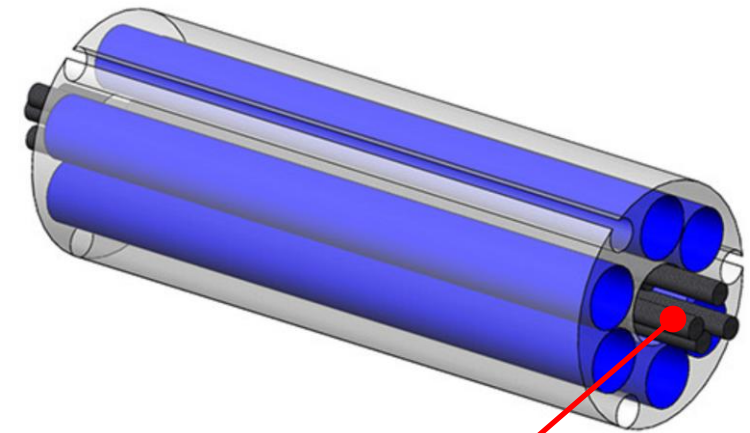
Brancadoro M, Manti M, Tognarelli S, Cianchetti M (2020) "Fiber Jamming Transition as a Stiffening Mechanism for Soft Robotics", Soft Robotics. Dec 2020. 663-674.



Applications - FJ

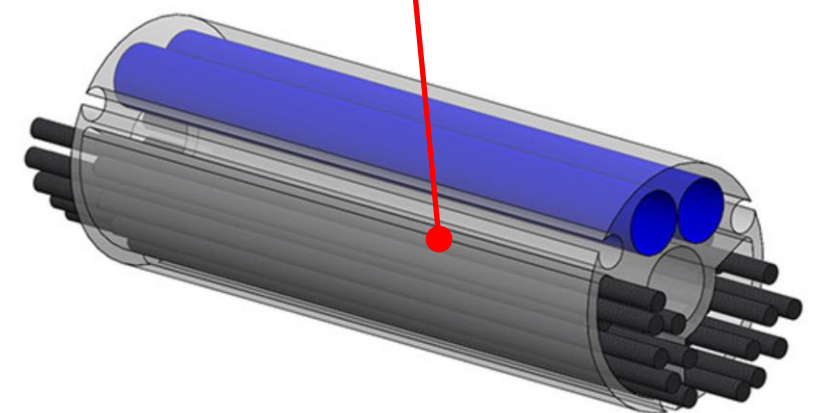


OPTION A



Fiber jamming system

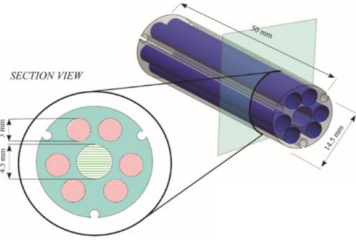
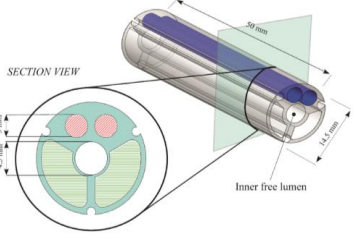
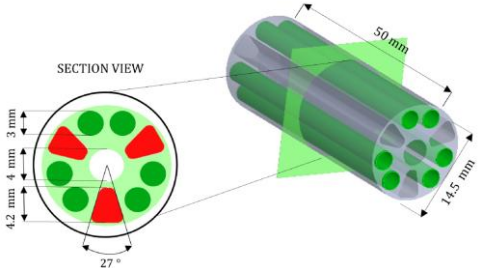
OPTION B



Brancadoro M, Manti M, Grani F, Tognarelli S, Menciassi A, Cianchetti M (2019) "Toward a variable stiffness surgical manipulator based on fiber jamming transition", Frontiers in Robotics and AI (section Soft Robotics), 6(12).



Applications - FJ

	WEIGHT	WORKING CHANNEL	PoM	VARIABLE STIFFNESS	STIFFENING/ STABILITY	WORKSPACE
	8.5 g	YES	4 (3 bending + elongation)	NO	-	+++
	9.2 g	NO 	4  (3 bending + elongation)	YES	+ 	++ 
	7.6 g	YES 	1  (bending)	YES	++ 	+ 
	9.1 g	YES 	4  (3 bending + elongation)	YES (+716%)	++ 	++ 

Brancadoro M, Manti M, Grani F, Tognarelli S, Mencias A, Cianchetti M (2019) "Toward a variable stiffness surgical manipulator based on fiber jamming transition", Frontiers in Robotics and AI (section Soft Robotics), 6(12).

Pagliarani N, Arleo L, Albini S, Cianchetti M (2023) "Variable Stiffness Technologies for Soft Robotics: A Comparative Approach for the STIFF-FLOP Manipulator", Actuators, 12(3):96.



Layer Jamming



Layer Jamming

Ambient pressure



- Filler free to slip
- High compliance

Vacuum



- Force exerted by ambient pressure
- Slip blocked
- Low compliance

Jamming system features



Cover

- deformable
- break resistant

Filler material

- low bending stiffness
- high tensile strength

M. Brancadoro, "Fiber Jamming Transition as a Stiffening Mechanism for Soft Robotics," soro.2019.0034

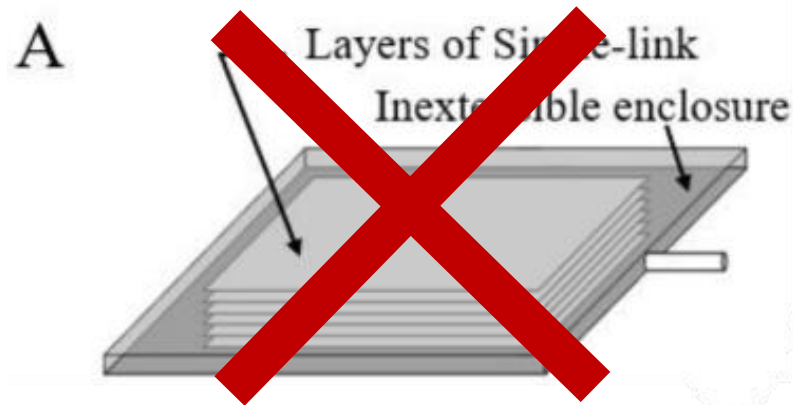
Y. S. Narang et Al, "Transforming the Dynamic Response of Robotic Structures and Systems Through Laminar Jamming," IEEE RA-L vol. 3, no. 2, pp. 688-695, April 2018



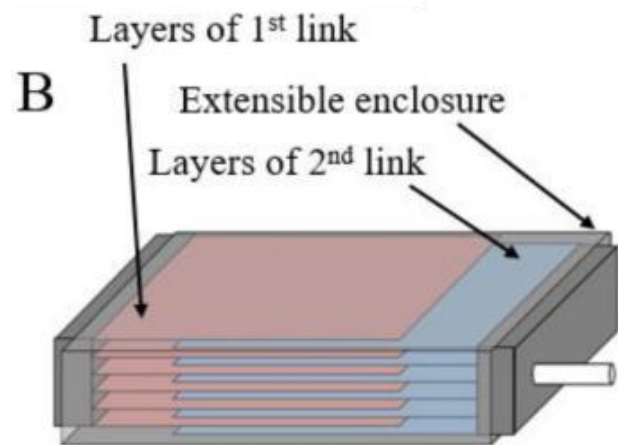
Axial forces: system behavior

Configuration

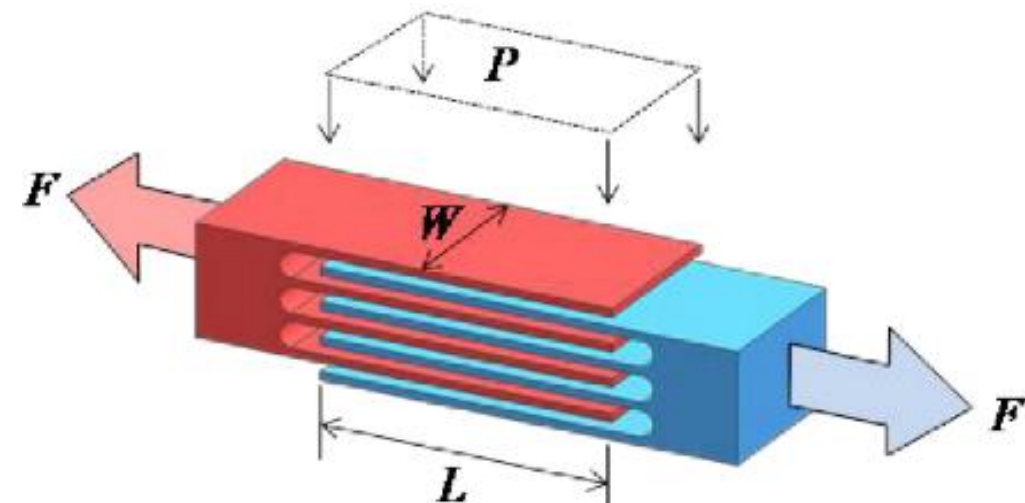
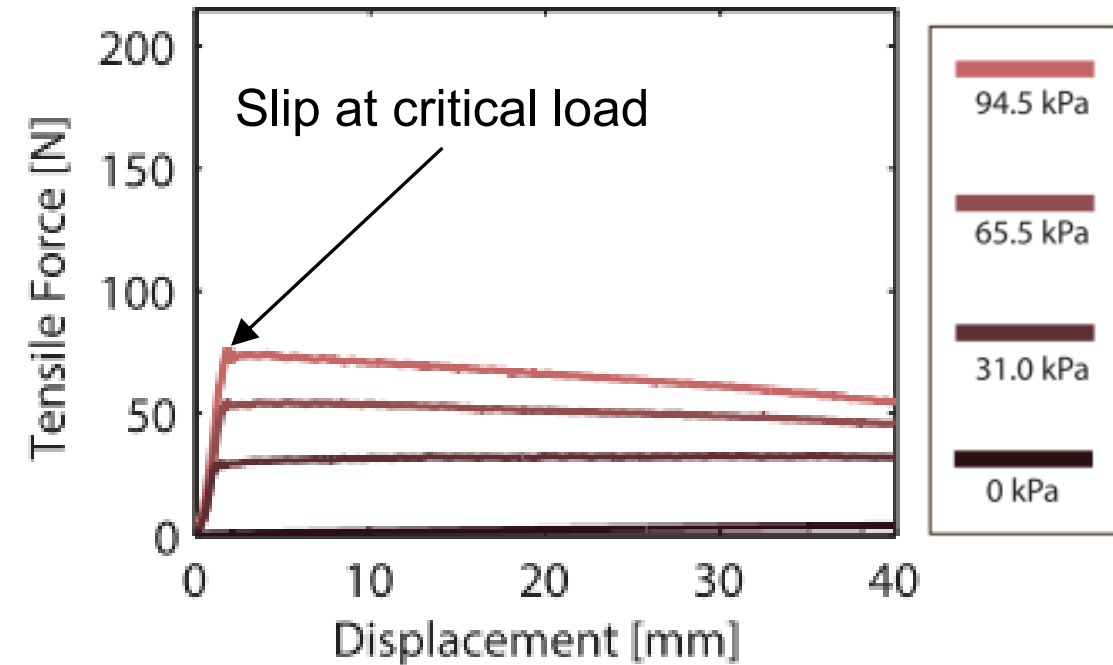
- (A) layers independent



- (B) layers grouped



Behavior



W. H. Choi et Al, "Soft, Multi-DoF, Variable Stiffness Mechanism Using Layer Jamming for Wearable Robots," 2019, doi: 10.1109/LRA.2019.2908493.

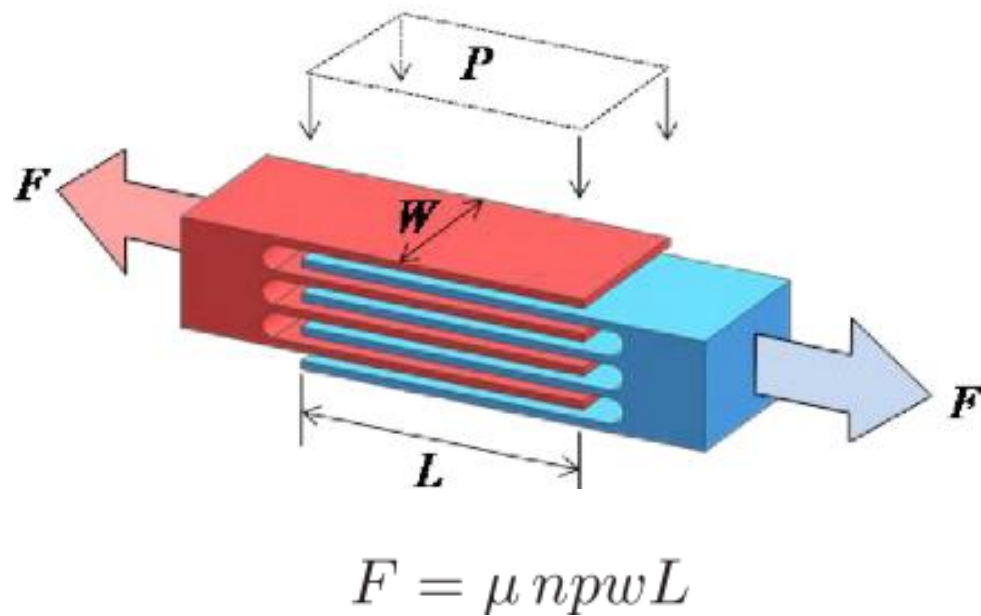
Choi,et Al, "A Soft, Controllable, High Force Density Linear Brake Utilizing Layer Jamming," 2018, doi: 10.1109/LRA.2017.2761938



Axial forces: models

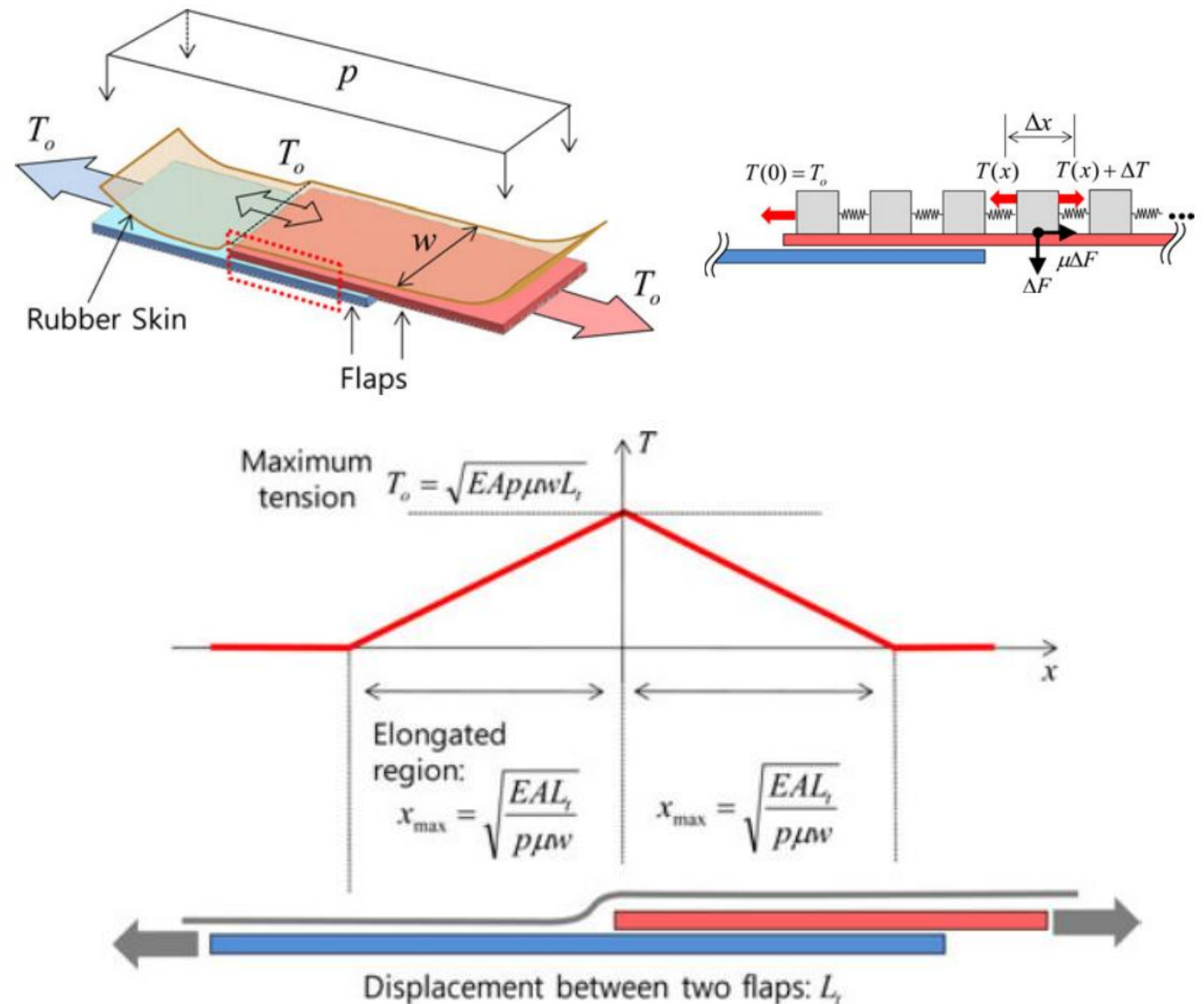
Basic concepts

Resulting force



- Resulting force is directly proportional to the size, friction coefficient and number of layers

Membrane tension

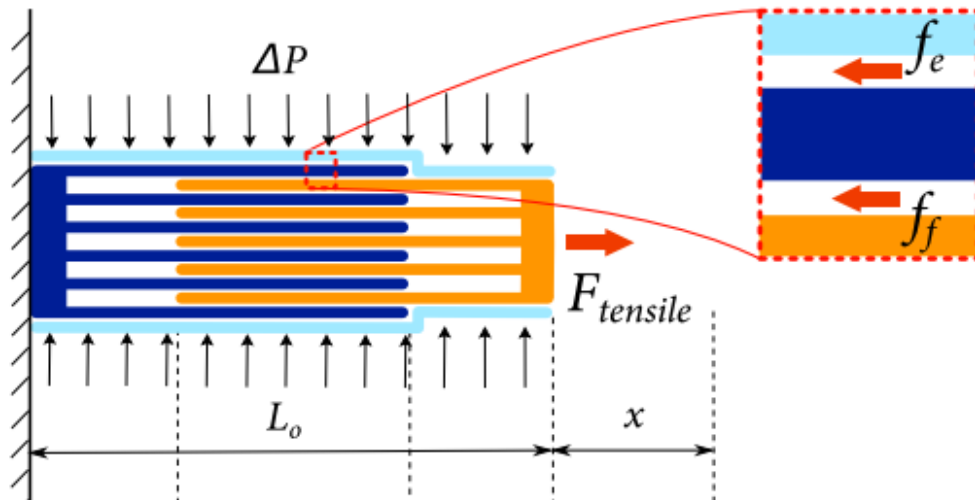


Y. -J. Kim et Al, "A Novel Layer Jamming Mechanism With Tunable Stiffness Capability for Minimally Invasive Surgery," IEEE T-RO, vol. 29, no. 4, pp. 1031-1042, Aug. 2013

Axial forces: models

Static conditions

Spring model



Pre-slip

$$F_{\text{tensile}} = k_{\text{effective}} x = (1/k_1 + 1/k_2 + 1/k_3)^{-1} x$$

$$k_i = E_l A_i / L_i (i = 1, 2, 3)$$

Post-slip

$$F = F_f + F_e$$

$$F_f = \mu(n-1)W(L_2 - x)\Delta P$$

$$F_e = 2\sqrt{E_e A_e \Delta P \mu_e W x}$$

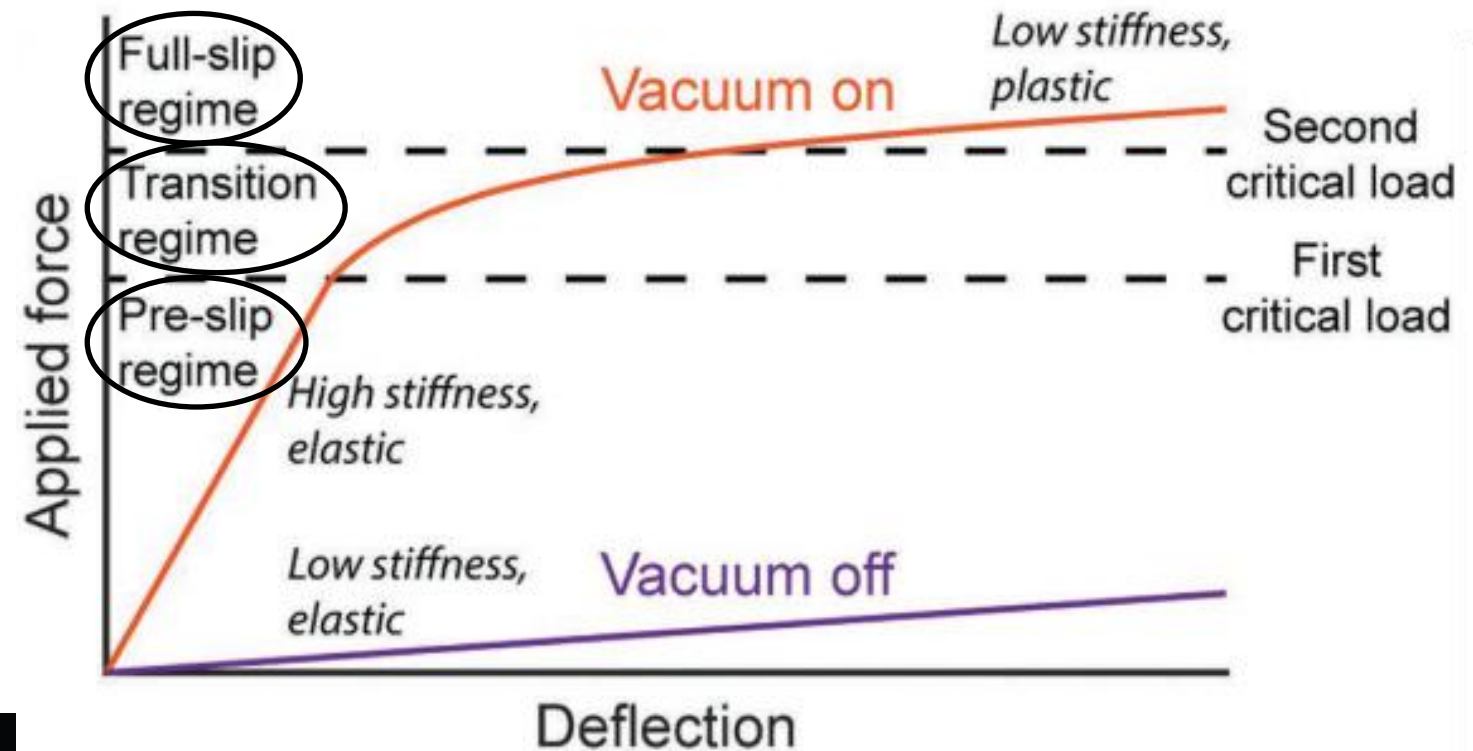
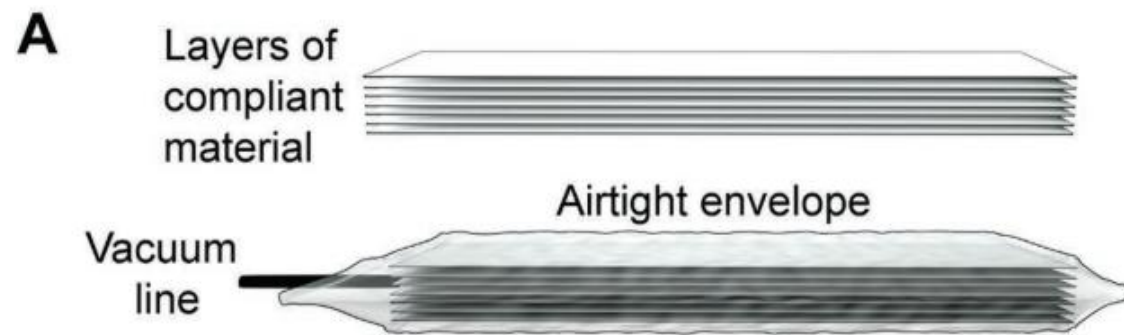
Dynamic conditions

$$E_{\text{dissipated}}(\Delta P) = \int F_{\text{tensile}}(x, \Delta P) dx$$

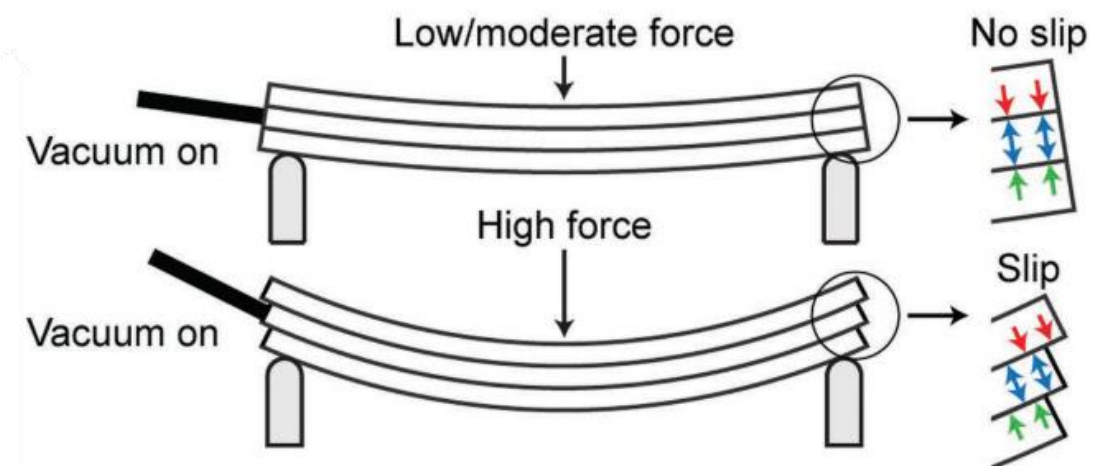
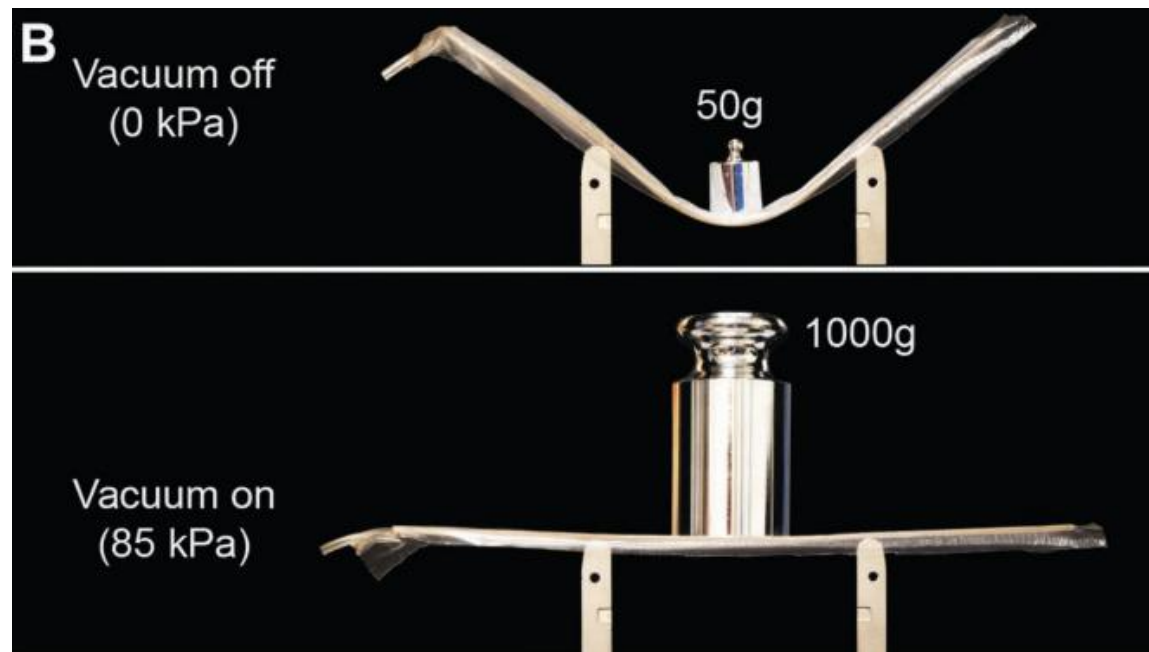


Transversal forces: System behavior

A-Layer Jamming system



B- 3 point bending configuration



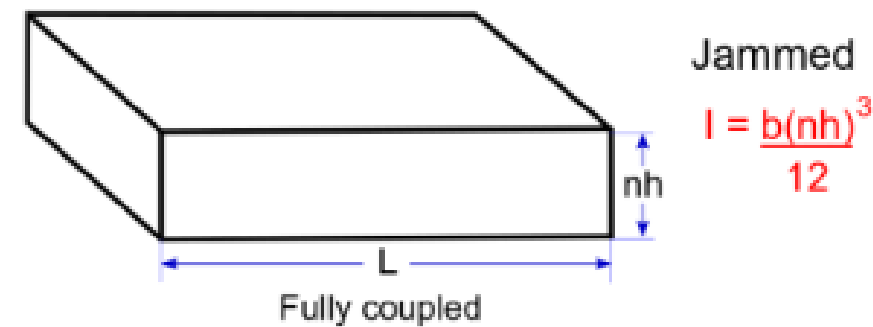
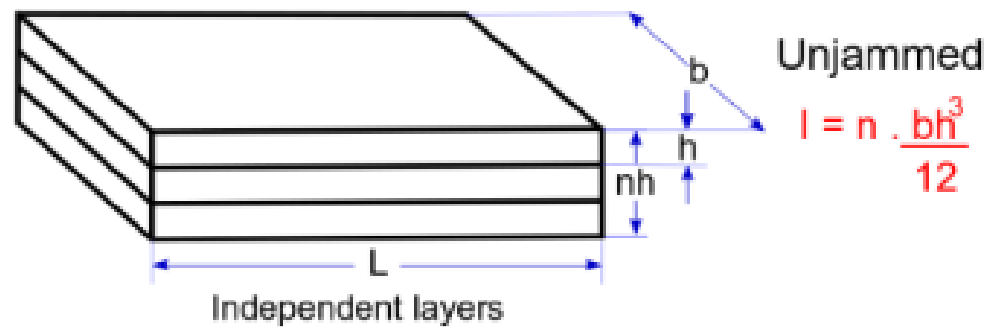
Narang, Y. S., Vlassak, J. J., Howe, R. D., Adv. Funct. Mater. 2018, 28, 1707136. <https://doi.org/10.1002/adfm.201707136>



Transversal forces: models

Static conditions

Analytical basic approach



$$\delta_{L/2} = \frac{FL^3}{48EI} \quad k = \frac{F}{\delta} = \frac{48EI}{L^3}$$

$$k_0 = \frac{4Ebnh^3}{L^3} \quad k_j = \frac{4Ebn^3h^3}{L^3}$$

$$r = \frac{k_j}{k_0} = n^2$$

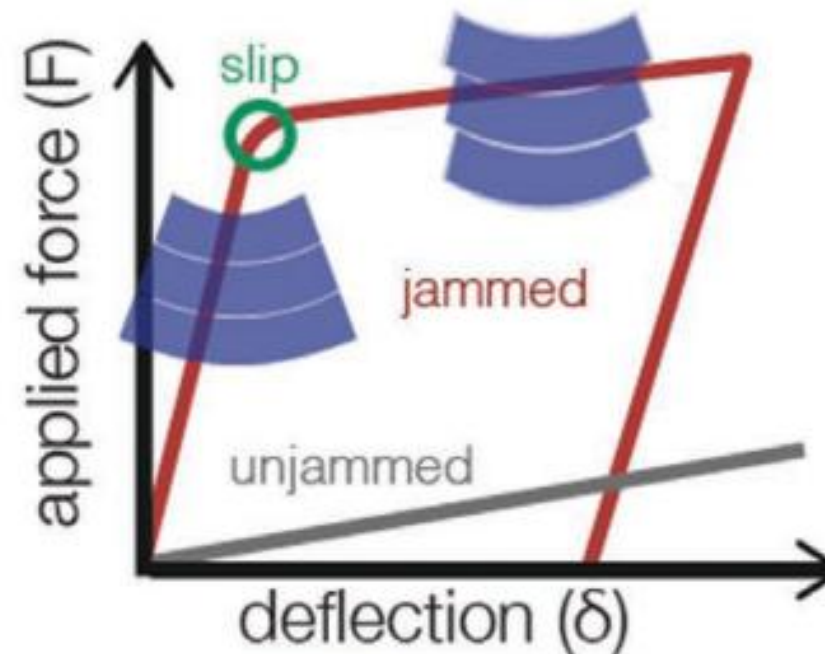
Slip force evaluation

$$\tau_{max} = \mu P = \frac{3F_{slip}}{2A}$$

max interlayer shear stress

From Jourawsky theory

$$F_{slip} = \frac{2\mu PA}{3}$$



Unjammed

$$F = k_0\delta$$

Jammed

$$F_{pre-slip} = k_j\delta$$

$$F_{slip} = 2\mu PA/3$$

$$F_{post-slip} = k_0\delta$$

Aktaş, B. et Al, "A Modeling Framework for Jamming Structures." Adv. Funct. Mater. 2021, 31, 2007554. <https://doi.org/10.1002/adfm.202007554>

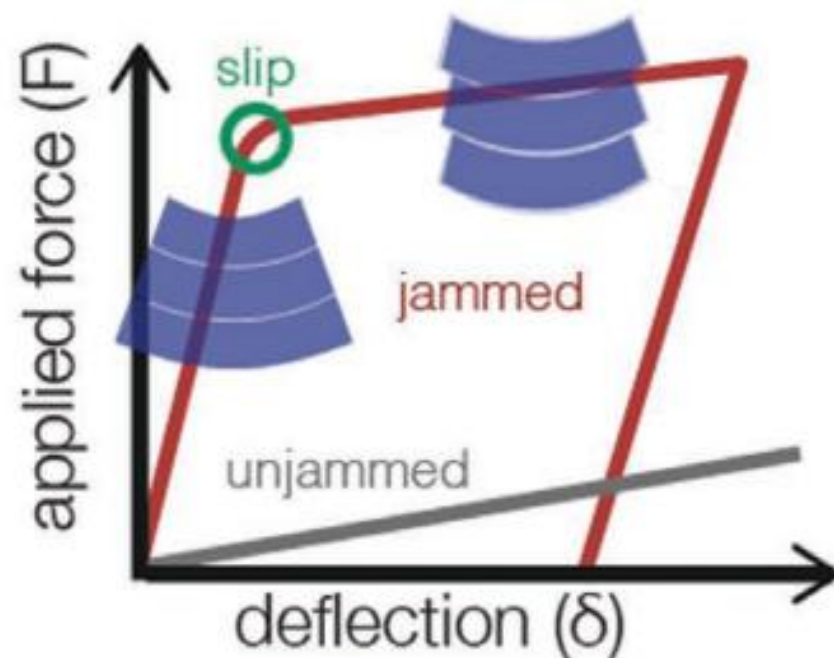


Transversal forces: models

Static conditions

Basic approach drawbacks

Model

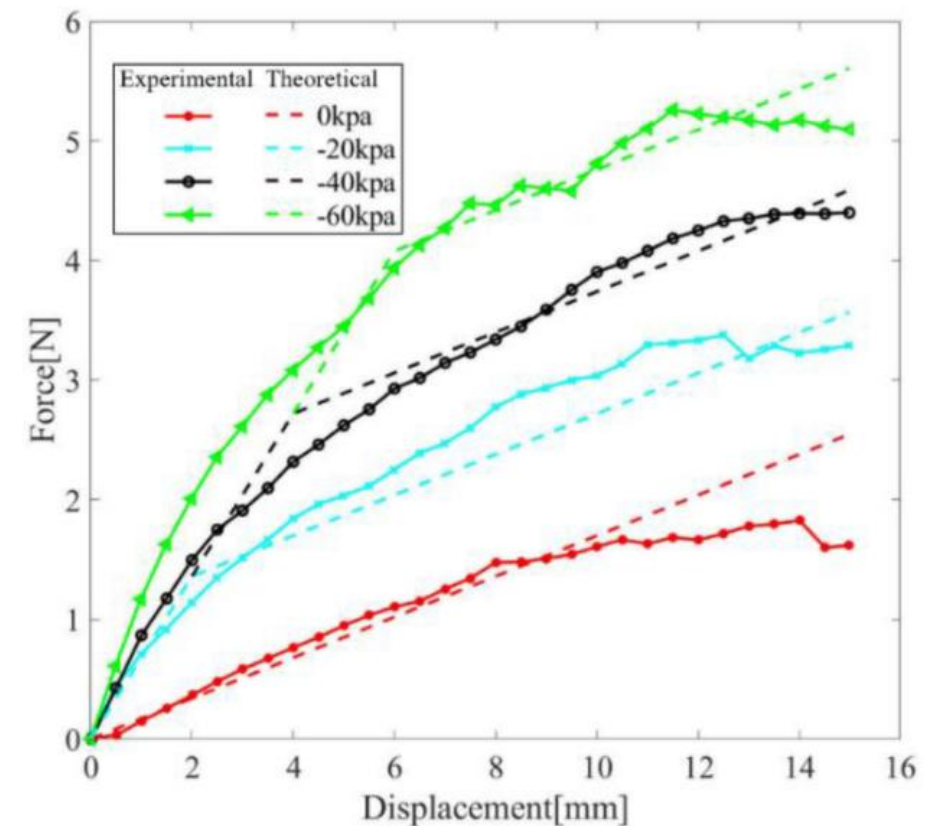


$$r = \frac{k_j}{k_0} = n^2$$

$$F_{slip} = \frac{2\mu PA}{3}$$

- Simplified behavior description
- Few factors affect the behavior

Actual behavior



- Stiffnesses increase with vacuum
- Non linear post-slip forces
- Transition region is always present



Transversal forces: models

Static conditions

Hybrid analytical–experimental model

1. Equilibrium equations

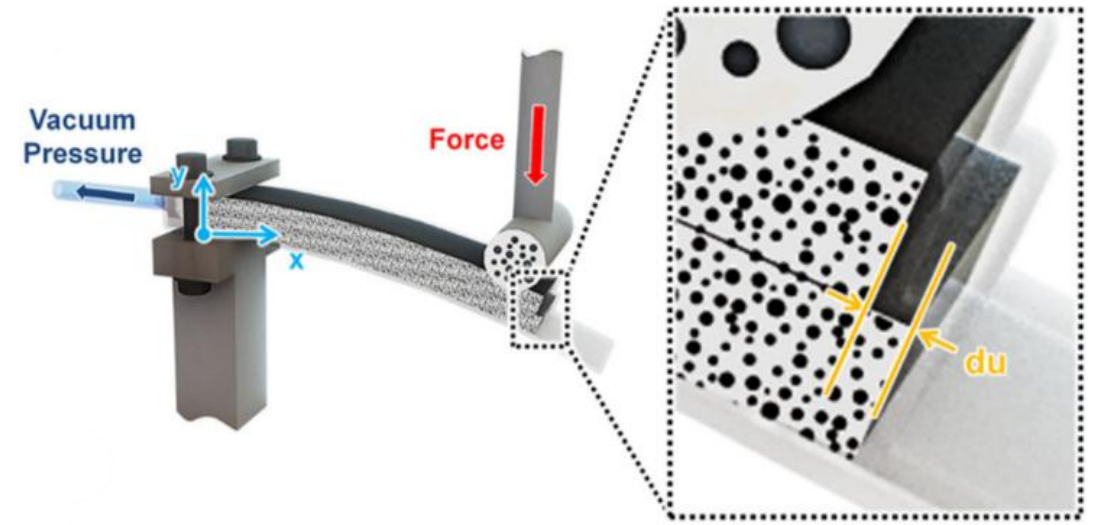
$$\sum F_x = \sum_{i=1}^N E_i b \left(\varepsilon_o t_i + \frac{1}{2} k t_i (t_i + 2y_i) \right) = 0$$



$$\sum M_z = \underbrace{M}_{\text{reference strain at the bottom face of the bottom layer}} + \sum_{i=1}^N E_i b \left\{ \frac{1}{2} (\varepsilon_o) [t_i^2 + 2t_i y_i] + \frac{1}{3} k [t_i^3 + 3t_i^2 y_i + 3t_i y_i^2] \right\} = 0$$

reference strain at the bottom face of the bottom layer

Curvature



$$\varepsilon_o = \frac{1}{\sum_{i=1}^N t_i} \left[- \sum_{i=1}^N \left(\frac{1}{2} k t_i (t_i + 2y_i) \right) + \sum_{i=1}^N t_i \Delta \varepsilon_i \right]$$

Actual Curvature

Actual discontinuities at each interface

Obtainable from du experimentally measured with Digital Image Correlation

2. Beam discretization

along x-axis

$$f_n = \frac{E_n}{E_{ref}} = \frac{\frac{M}{E_{ref} b}}{\frac{M}{E_n b}} = \frac{\frac{M}{E_{ref} b}}{\sum_{i=1}^N \left\{ \frac{1}{2} (\varepsilon_o - \Delta \varepsilon_i) [t_i^2 + 2t_i y_i] + \frac{1}{3} k [t_i^3 + 3t_i^2 y_i + 3t_i y_i^2] \right\}}$$

Acevedo, 2020, Characterization and Modeling of Layer Jamming for Designing Engineering Materials with Programmable Elastic-Plastic Behavior



Transversal forces: models

Static conditions

Hybrid analytical–experimental model

3. Link between f_n and $\frac{v}{v_{ref}}$

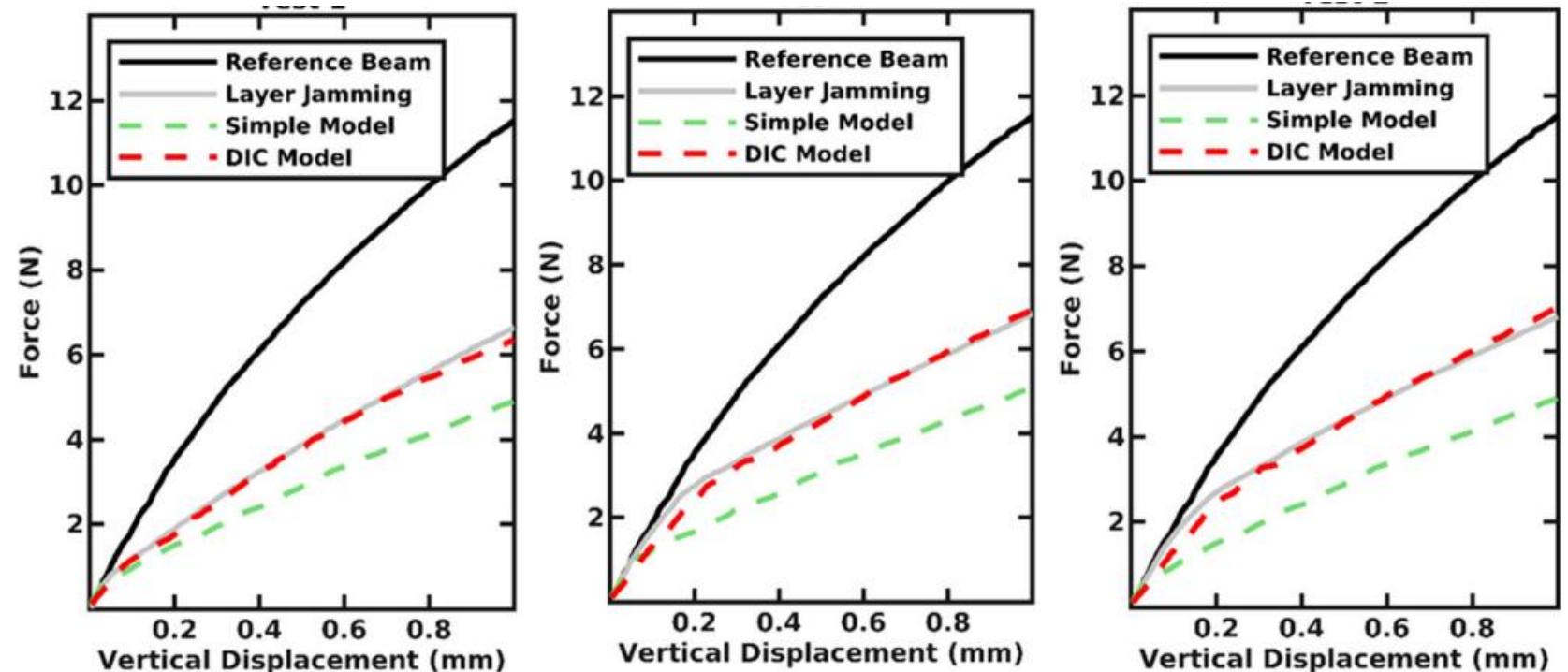
Actual deflection

Deflection of the coupled configuration

$$v = \frac{v_{ref}}{N_s^3} \sum_{n=1}^{N_s} \left[\frac{1 + 3N_s \left(1 - \frac{x_n}{L}\right) \left(N_s \left(1 - \frac{x_n}{L}\right) + 1\right)}{f_n} \right]$$

Validation

- Non linearity taken into account
- Accurate prediction

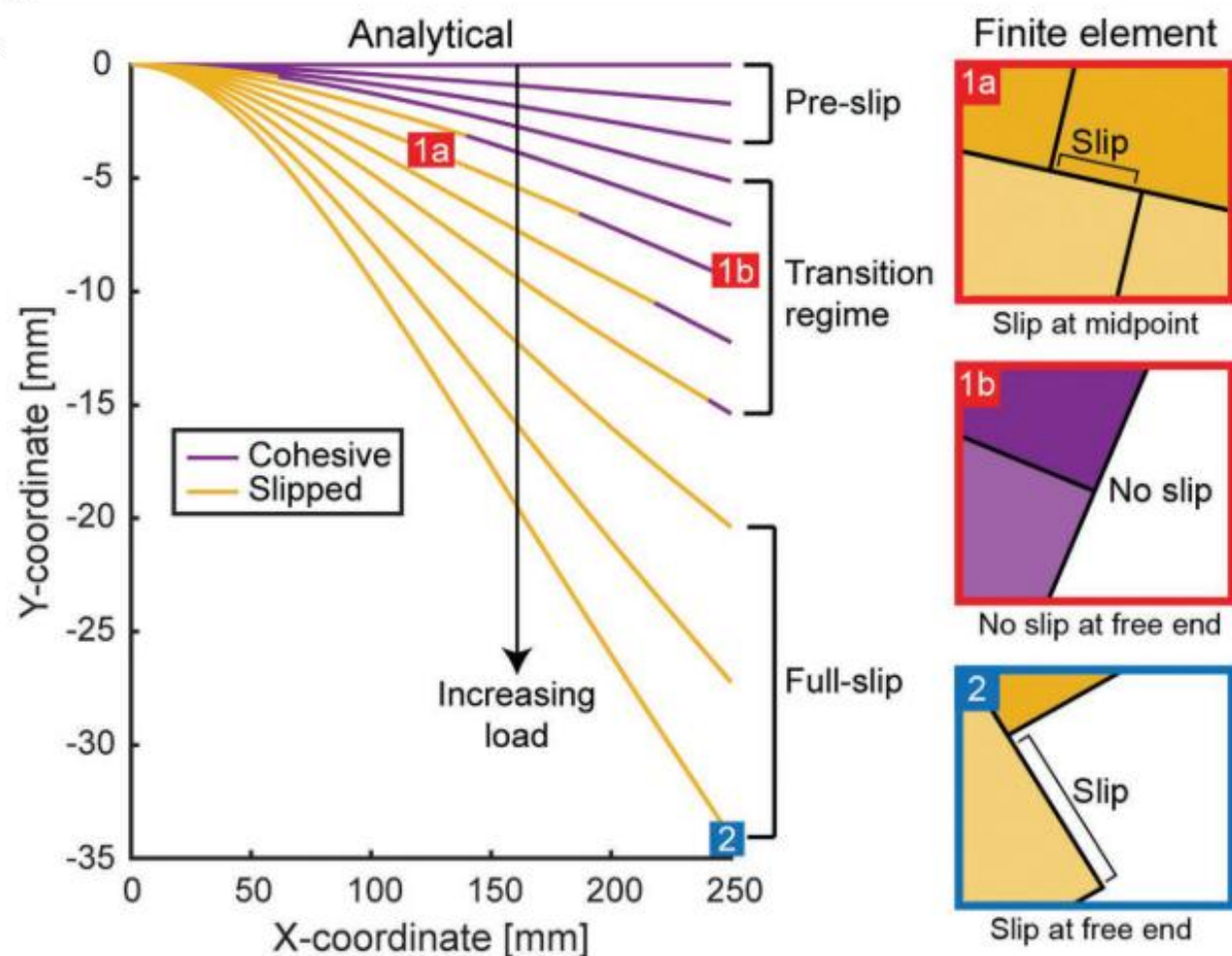


Transversal forces: models

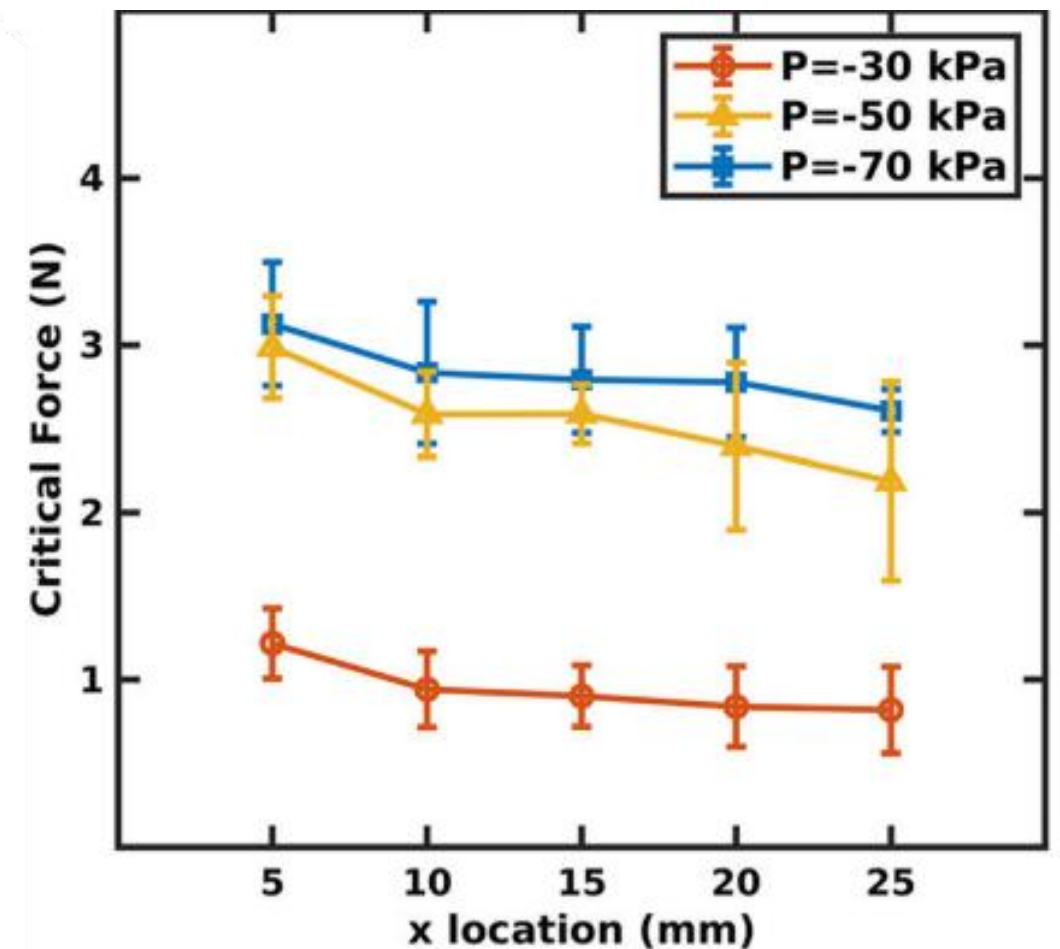
Static conditions

Slip evolution along x

- Dissipated energy based analytical model in (1)



- Experimental measures in (2)

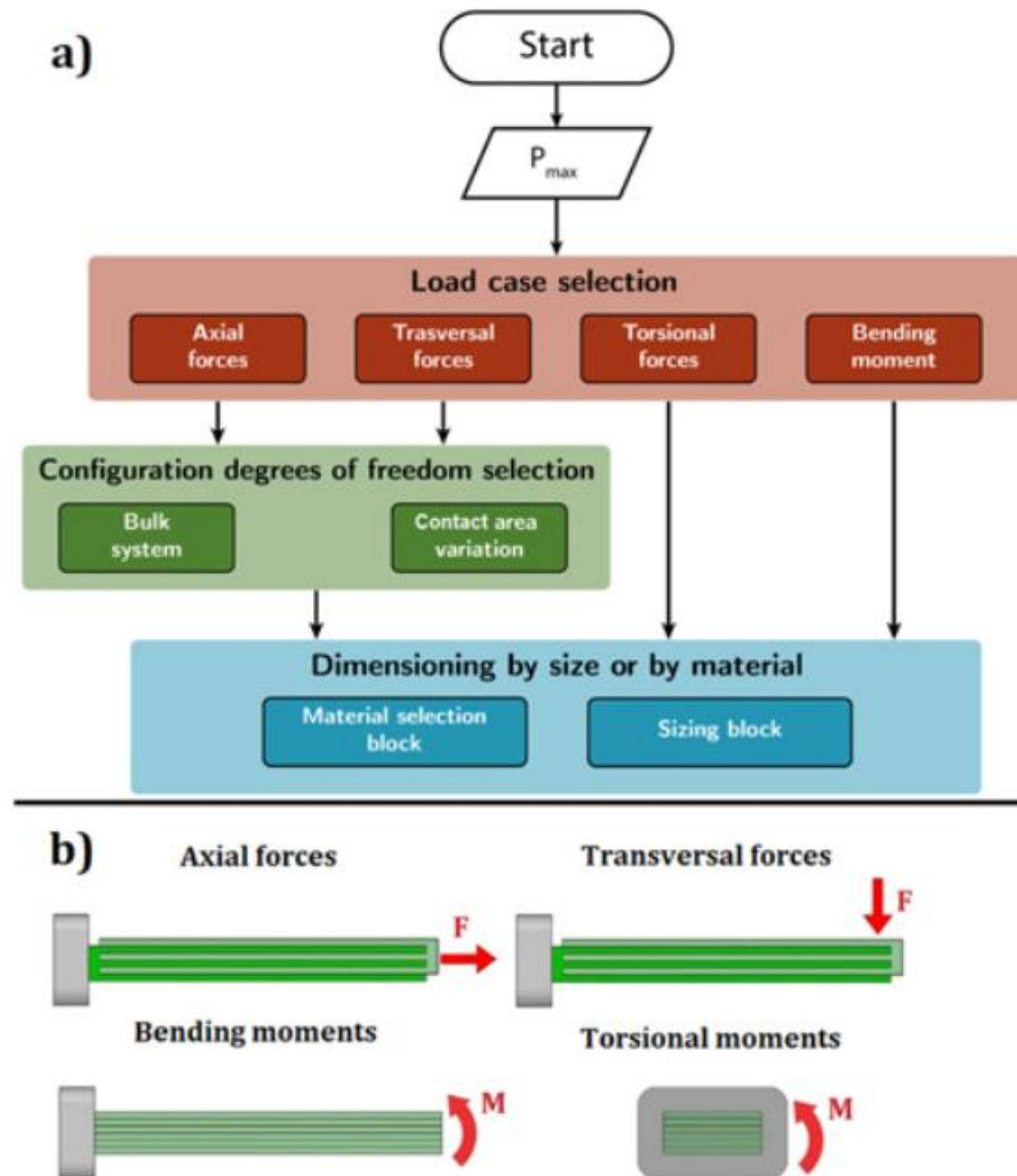


(1) Narang, Y. S., Vlassak, J. J., Howe, R. D., Adv. Funct. Mater. 2018, 28, 1707136.

(2) Acevedo, 2020, Characterization and Modeling of Layer Jamming for Designing Engineering Materials with Programmable Elastic-Plastic Behavior



Design Methodology for Layer Jamming



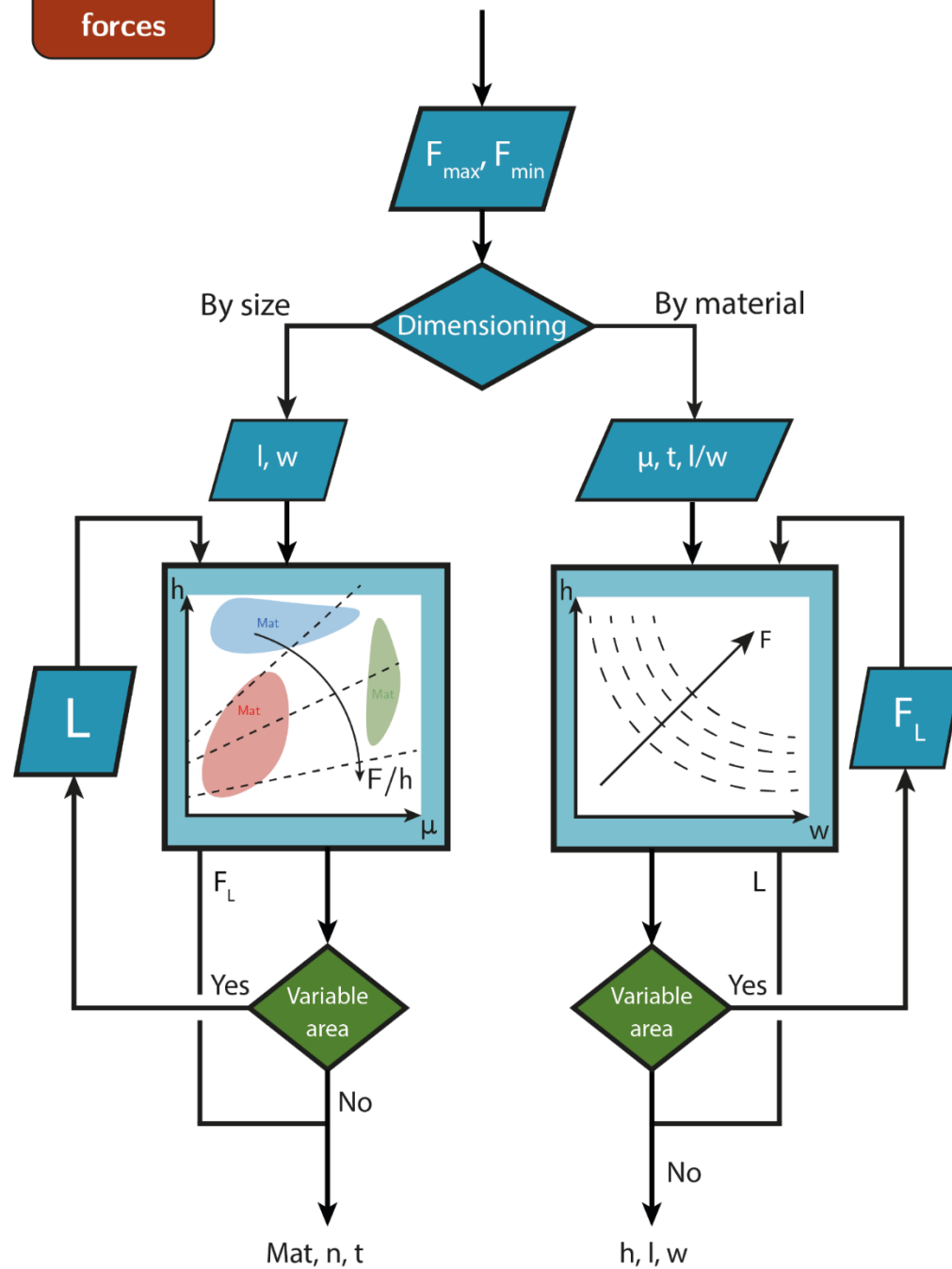
- 4 step design process:
 - i) Vacuum level
 - ii) load case between 4 possibilities
 - iii) A or B design type
 - iv) known terms definition
- Load case selection
 1. Axial forces
 2. Transversal forces
 3. Bending moments
 4. Torsional moments



Design Methodology for Layer Jamming

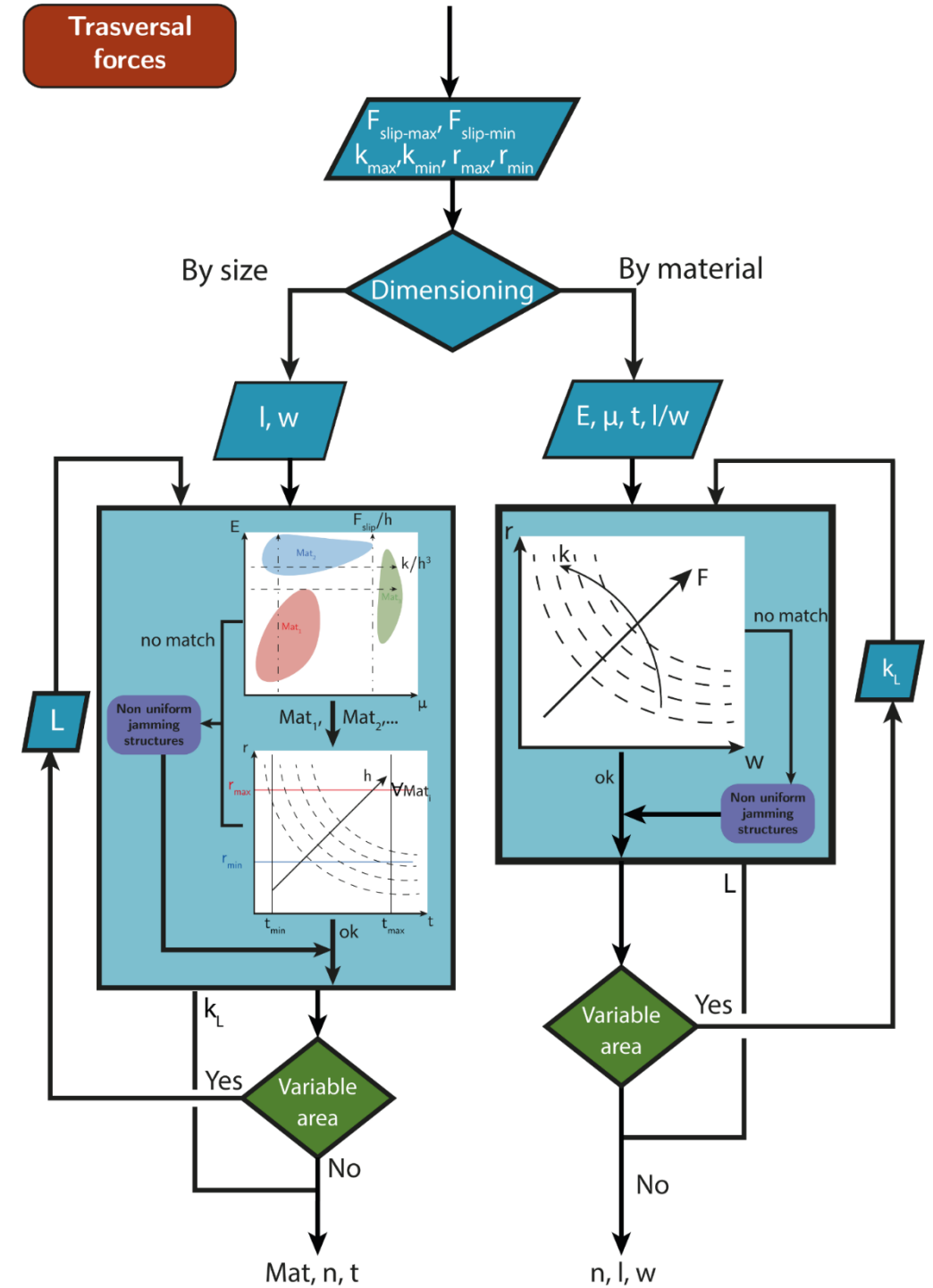
1.

Axial forces



2.

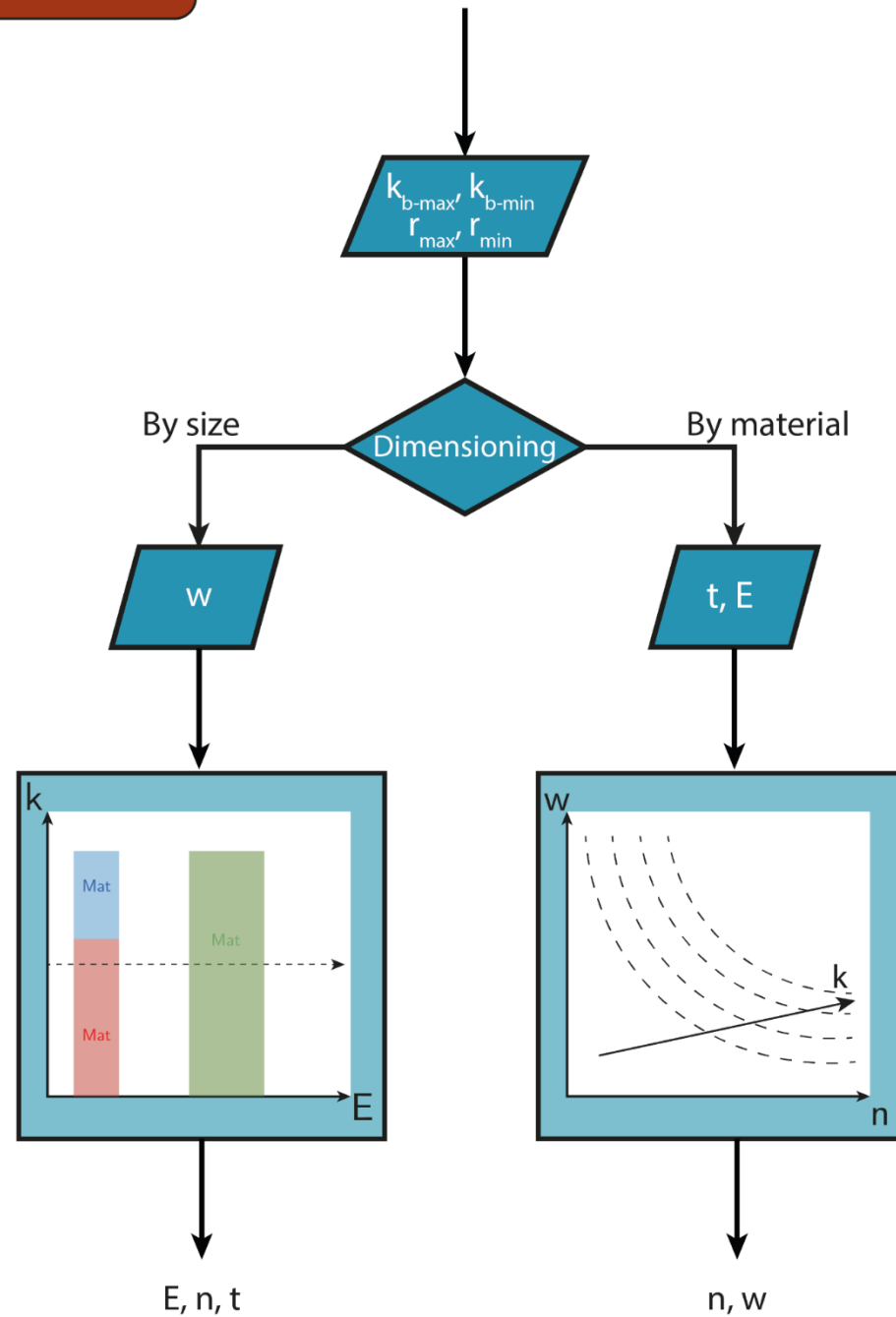
Trasversal forces



Design Methodology for Layer Jamming

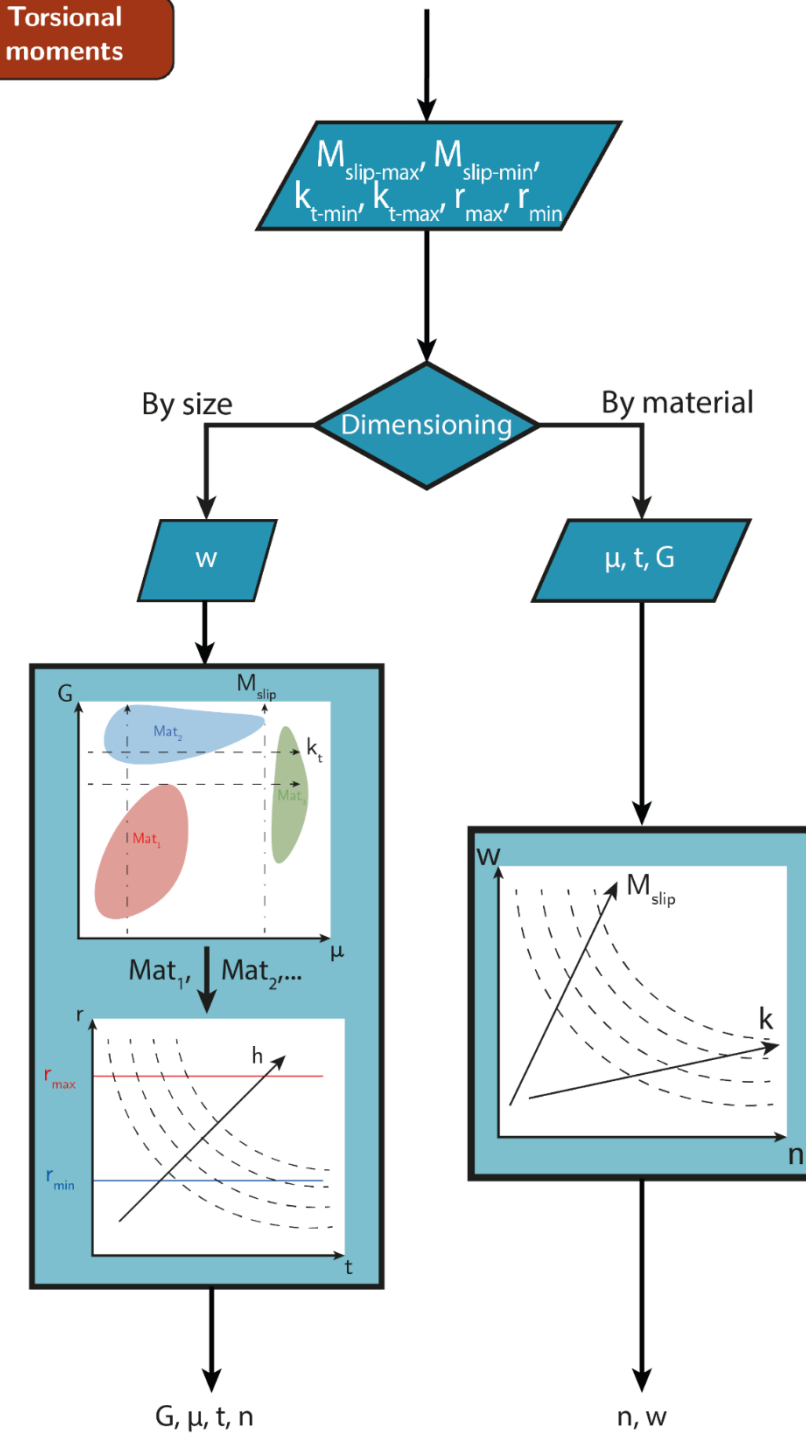
3.

Bending moments

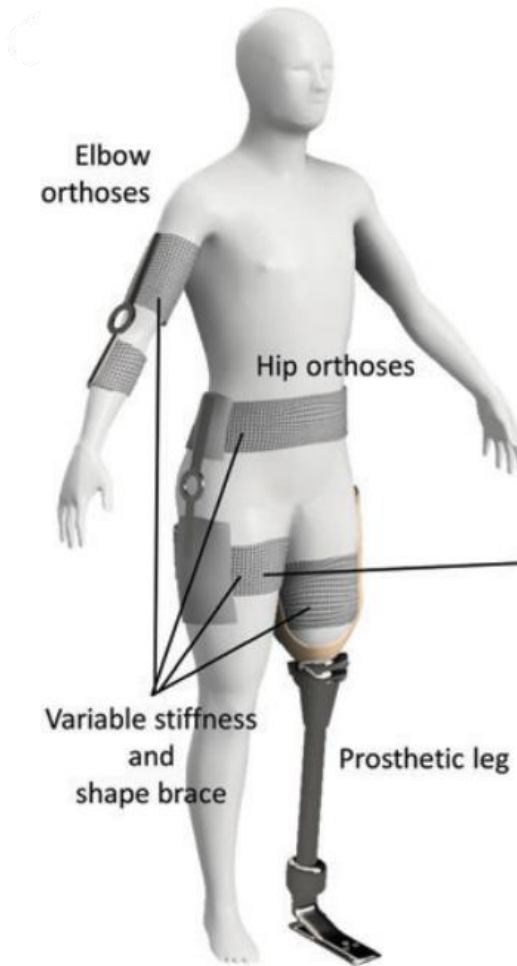
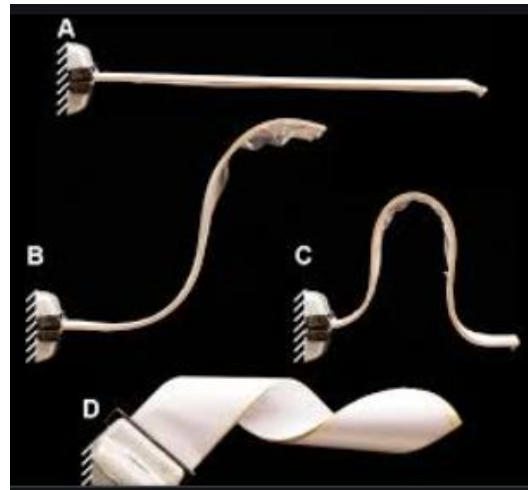


4.

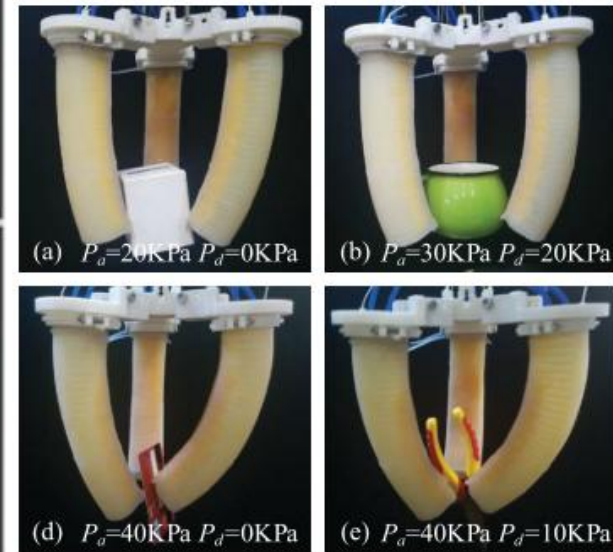
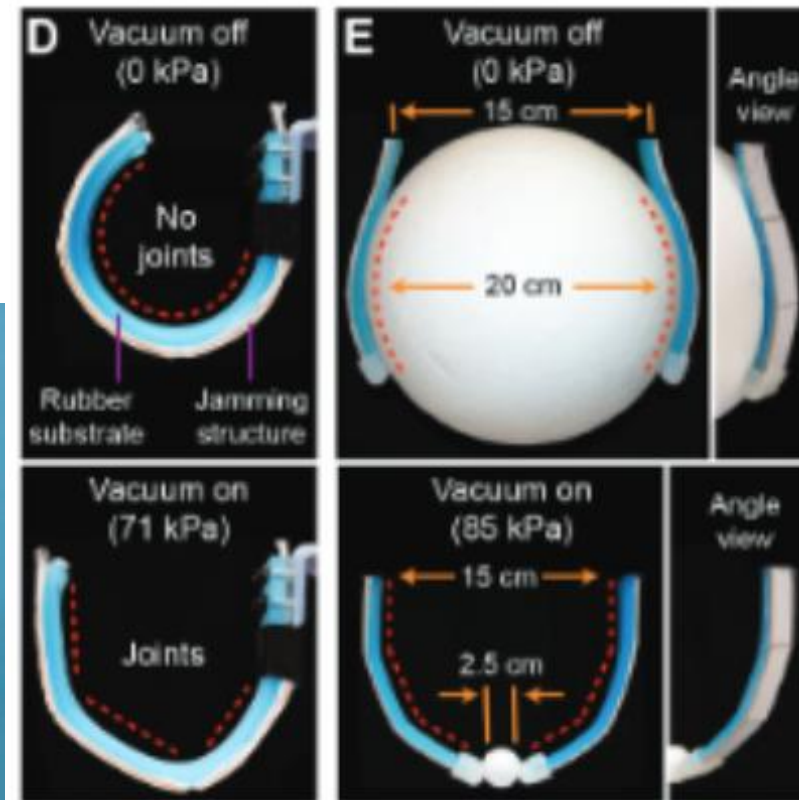
Torsional moments



Applications - LJ



Jamsheets



M.Ibrahimi et Al, "A Layer Jamming Actuator for Tunable Stiffness and Shape-Changing Devices", SoRo 202, 18:1, 85-96

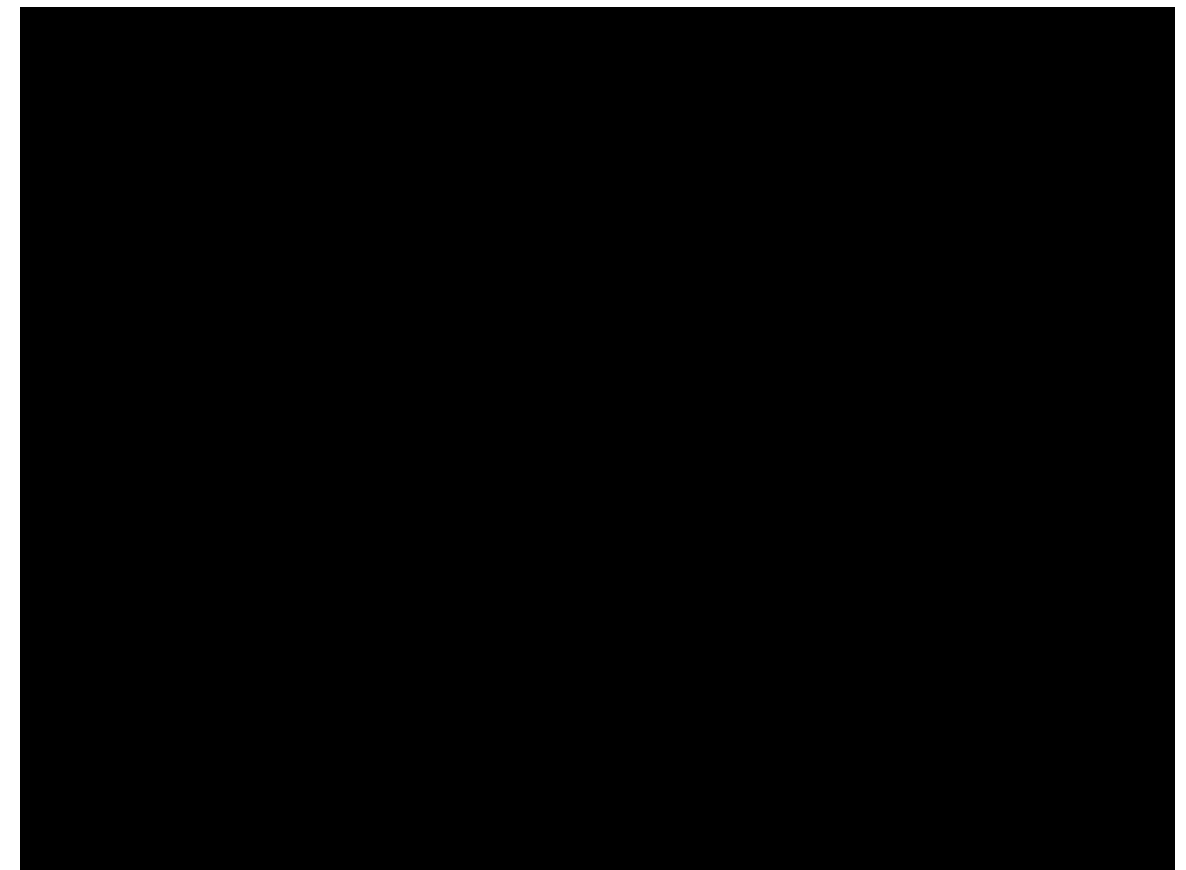
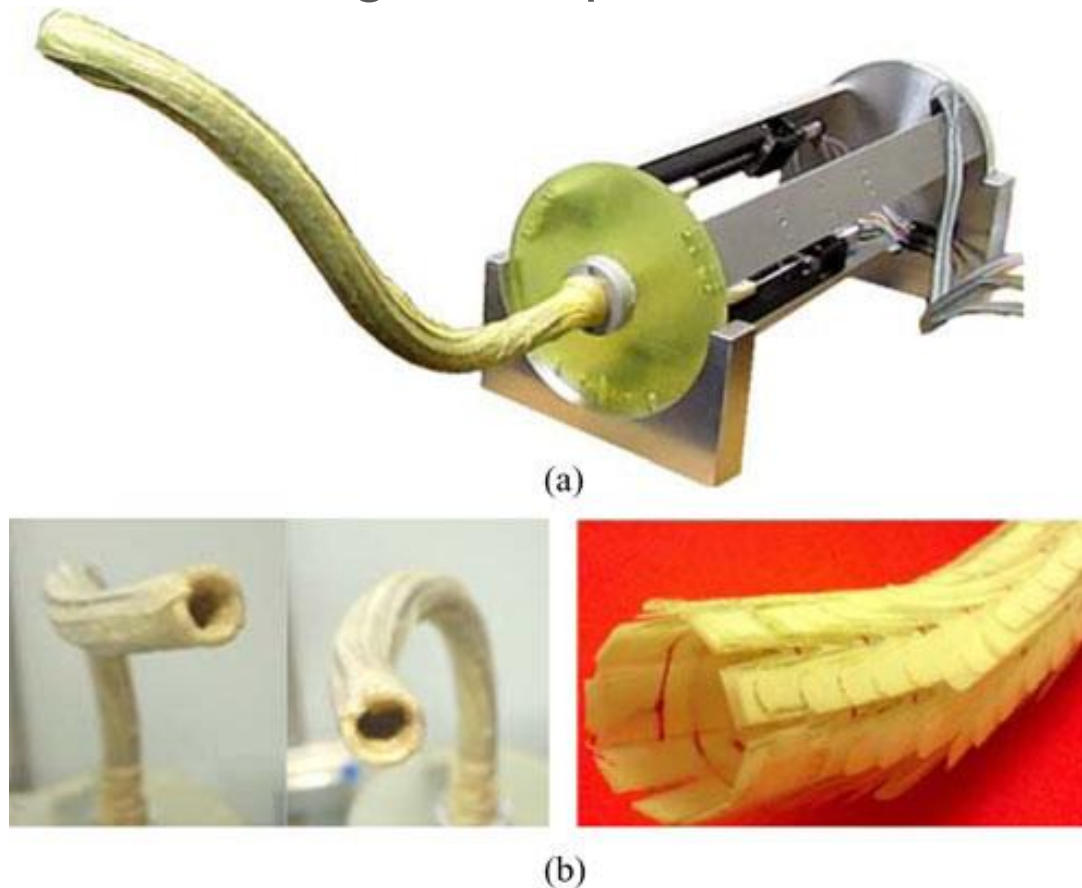
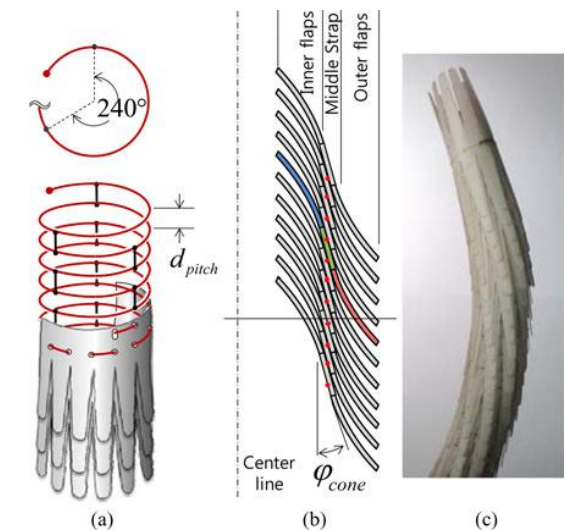
Pei Jiang et al 2019 Bioinspir. Biomim. 14 036009



Applications - LJ

Snake-like manipulator

The layer jamming mechanism exploits the friction present between layers of thin material, which can be controlled by a confining pressure. Due to the mechanism's hollow geometry, compact size, and light weight, it is well suited for various minimally invasive surgery applications, where stiffness change is required.



Kim, Y.-J.; Cheng, S.; Kim, S. & Iagnemma, K. A Novel Layer Jamming Mechanism With Tunable Stiffness Capability for Minimally Invasive Surgery Robotics, IEEE Transactions on, 2013, 29, 1031-1042

